Energy Economics 155 (2026) 109176

Contents lists available at ScienceDirect

Energy Economics

journal homepage: www.elsevier.com/locate/eneco

Deep-learning-based optimal auction design in electricity markets

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ARTICLE INFO

JEL classification:
C45
D44
D47
Q41
Q42

Keywords:
Electricity markets
Optimal auction design
Deep learning
Renewable power integration

ABSTRACT

Auctions are widely used in electricity markets as a mechanism for central operators to ensure demand is satisfied in a cost-effective manner. Recently, the *RegretNet* framework has been proposed to tackle the optimal auction design problem with a deep learning approach. In this paper, we extend this framework to discover nearly optimal designs for electricity auctions. This is achieved by: (i) altering the neural network architecture to determine the *number of units* to allocate and to incorporate *demand constraints*; (ii) representing the information rent as part of the generator's cost to capture *individual rationality*; (iii) introducing unbounded profit functions to handle capacity constrained generators; (iv) relaxing the learning problem to handle the *capacity* and *incentive compatibility* constraints; and (v) augmenting the constraints in the learning problem to handle correlated unit-costs. These extensions enable us to consider: (i) uncertain capacity and demand, possibly due to supply failures or wind–solar integration; (ii) correlated unit costs, caused by seasonal effects or shocks; and (iii) heterogeneous multi-time slot dispatch, capturing time-varying generation costs. Through experimentation we demonstrate that the method is able to recover known analytical solutions, achieving precise cost-level approximations (with errors $< 1\%$) and minimal constraint violations (≤ 0.001). Finally, we employ the method to assess the effect of renewable power integration in the Colombian wholesale electricity market. These results highlight the ability of the method to support the design of electricity markets considering the technical characteristics of the generators, the uncertainty around their capacities and costs, as well as their strategic behavior.

1. Introduction

Electricity is a fundamental component of modern societies, with significant environmental, welfare, and economic implications. The need for affordable, reliable, and clean energy has led to the continual improvement in the design of electricity markets (Chu and Majumdar, 2012; Parag and Sovacool, 2016; Cramton, 2017). Centrally committed markets have historically been organized as multi-unit auctions, where power generators compete to supply electricity units and meet load demand (Sioshansi et al., 2008; Wolfram, 1997). An auction design is defined by its allocation and payment rules, which determine how much electricity each generator must dispatch and the payment it receives, respectively. In this context, the goal is to design an auction that minimizes expected generation costs, while satisfying electricity demand and adhering to the generators' capacity constraints. Moreover, as generators aim to maximize profits by strategically deciding whether to participate in the auction and the unit price to offer, the auction design problem focuses on strategy-proof auctions that incentivize participation (*individual rationality*) and truthful bidding (*incentive compatibility*) (Hobbs et al., 2000b; Léautier, 2001). As a result, optimal auctions provide a baseline to evaluate the performance of existing mechanisms and serve as a guide for regulation, offering insights into the auction design problem and assessing the impact of policy interventions.

To tackle this problem, the optimal auction design literature, initiated by the work of Myerson (1981), has provided analytical solutions to this problem, albeit limited to relatively simple scenarios. Additionally, the scalability of linear programming approaches proposed by the automated design literature using off-the-shelf solvers has proven to be computationally expensive (Guo and Conitzer, 2010). Recently, Dütting et al. (2019) proposed the *RegretNet* framework to find nearly optimal designs for revenue-maximizing multi-item auctions using deep learning. This framework leverages neural networks to compute allocation and payment rules based on bidders' valuations and casts the non-convex design optimization problem as a constrained learning task. This method eliminates the need to discretize the bidders type space (e.g., according to their unit costs), reducing the computation times needed to find optimal auctions considerably.

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https://doi.org/10.1016/j.eneco.2026.109176
Received 20 December 2024; Received in revised form 16 December 2025; Accepted 30 January 2026
Available online 10 February 2026

In this work, we adapt and extend the *RegretNet* framework (Dütting et al., 2019), to discover near-optimal designs for electricity auctions that minimize expected generation costs and incentivize truthful bidding in simplified settings. In particular, we extend the framework to *multi-unit* auctions with capacity constraints for procuring electricity demand over multiple time slots (e.g., hours) in a day. This is achieved by five key extensions: (i) the neural network architecture is modified to model the allocation rule as the number of units allocated (instead of the probability of assigning each specific item as in the multi-item setting), and to incorporate the *demand constraint*; (ii) the *individual rationality* condition for procurement auctions is introduced within the payment rule by first estimating a non-negative component that represents the information rent (i.e., the extra payment needed to incentivize generators to report their true unit costs), and adding it to the generator's cost; (iii) unbounded profit functions are used to handle capacity constrained generators; (iv) relaxations are included in the learning problem to handle *both* the *capacity* and *incentive compatibility* constraints; and (v) the constraints in the learning (optimization) problem are augmented to handle the case where the unit-cost functions are correlated across suppliers. These extensions enable us to consider key features of electricity auctions, namely:

(1) Uncertain capacity and demand arising from noise, supply failures, and wind–solar integration.
(2) Correlated unit costs across generators stemming from seasonal and fuel price variations.
(3) Multi-time slot dispatch with non-identical electricity units across time slots.

To assess the proposed method, we conduct many experiments, starting with a single time slot, independently drawn unit costs, and constant capacities to demonstrate that the method can recover known analytical solutions (derived from Iyengar and Kumar (2008) and Chaturvedi (2015)), achieving precise cost-level approximations (with errors $< 1\%$) and minimal constraint violations (≤ 0.001). Next, we show the versatility of the method to discover optimal auctions in settings with uncertain capacity and demand, correlated costs, and varying unit costs between time slots. Finally, we conduct a case study based on real data from the Colombian electricity market to assess the effect of wind and solar power integration on generation costs. The results show that wind and solar power integration is slightly more effective at reducing *average* costs than increasing hydropower capacity, but it is significantly more effective to reduce the cost *high percentiles*. In fact, it lowers the risk of *high generation costs* during low rainfall periods, achieving reductions in the 95th percentile of 8.8%, 8.2%, and 16.1% at 10%, 20%, and 30% capacity expansions, respectively. These results show the ability of the extended framework to provide policy recommendations on the design of electricity markets.

1.1. State of the art

In designing electricity auctions, technical considerations, such as uncertain capacity and demand, intertemporal dispatch decisions, and diverse generation technologies add complexity to the analysis. In addition, the design should balance other competing objectives beyond cost minimization, including price stability, operational reliability, and environmental sustainability. For instance, integrating renewable energy promises to reduce greenhouse gas emissions and offer cost-effective alternatives to conventional energy sources thanks to recent technological advancements (Panwar et al., 2011; Ang et al., 2022). However, the decentralized and uncertain nature of wind and solar energy poses difficulties in managing operational risk and ensuring consistent energy availability throughout the day, often requiring innovative solutions, such as storage systems and demand response programs (Chu and Majumdar, 2012; Aghaei and Alizadeh, 2013; Wang et al., 2022).

Furthermore, in the energy economics literature, the focus on optimal designs has been relatively limited compared to the extensive work on characterizing equilibria of electricity auctions (mainly uniform and pay-as-bid auctions). Examples include Fabra et al. (2006) on pay-as-bid and uniform-price auctions, Reguant (2014) on auctions with complex bids accounting for non-convex cost functions, and Fabra and Llobet (2023) on settings with privately known capacities. On the other hand, works in the operations research literature have focused on power system optimization for optimal electricity generation dispatch. Widely-studied problems include the unit commitment problem, which determines the optimal ON/OFF schedule for thermal generators to minimize operating costs, and the economic dispatch problem, which decides the optimal electricity output for each committed generator and the single unit electricity price to meet demand (Cohen and Sherkat, 1987; Padhy, 2004). However, these problems often assume generators bid their true unit costs, treating them as price takers. This assumption can be inaccurate, especially in markets with transmission constraints, capacity concentration, inelastic demand, and power generation influenced by seasonal variations (Pereira et al., 2005; McRae and Wolak, 2017; Balat et al., 2023).

While previous works have addressed strategic bidding by integrating generators' first-order optimality conditions into the programs and developing methods to handle these non-convexities (Hobbs et al., 2000a; Pereira et al., 2005), most recent works have focused on extending these problems to better reflect current electricity system features and needs, incorporating renewable generation, battery storage strategies, demand response programs, and microgrids (Hetzer et al., 2008; Atwa et al., 2009; Mazidi et al., 2014), but without incorporating the bidders' strategic behavior.

While the *RegretNet* framework proposed by Dütting et al. (2019) is a promising tool to find nearly optimal auction designs, it was developed for revenue-maximizing multi-item auctions, instead of the procurement auctions found in electricity markets. Previous extensions to this framework, and closely related to this paper, include Feng et al. (2018), which considered multi-unit auctions involving bidders with private budgets, relaxing the quasilinear preferences assumption. Also, Bhardwaj et al. (2023) modeled auctions for the procurement of homogeneous units with volume discounts by defining the allocation rule as the *number* of units to allocate to each bidder, rather than specifying *which* units are allocated. As of now, this deep learning approach has not considered capacity constrained bidders, which is a key feature in electricity auctions, and its extensions for procurement multi-unit auctions remain limited.

1.2. Main contributions

First, to the best of the authors' knowledge, this work is the first to introduce specific features from electricity auctions to discover optimal designs by extending the *RegretNet* framework. Said features include uncertain capacity and demand, correlated unit costs across generators, and non-identical multi-time slot dispatch. Our approach successfully recovers optimal auction designs for settings where bidders are capacity constrained, and finds new mechanisms for electricity auctions that incorporate the previously mentioned features. Secondly, it provides guidance on how various assumptions and policies influence the optimal design and the generation cost levels and stability. Overall, this study serves as a bridge between the electricity auction and the optimal auction design literature through the deep learning approach, and hopes to contribute to the relatively recent research field at the intersection between machine learning techniques, economic analysis and operations research.

Moreover, we believe this work provides a flexible framework for tackling many more applications for electricity and procurement auctions overall, such as contamination and fairness constraints, multi-part and block bidding, reserve and battery storage management, demand response programs, among others.

Outline. The remainder of this document is organized as follows: Section 2 introduces the auction design learning problem, Section 3 describes the model's architecture and training algorithm, Section 4 presents experimental results, and Section 5 concludes and proposes further extensions.

Notation. In the document, vectors and matrices are written in **bold**. Let $\mathbf{0}$ and $\mathbf{1}$ denote vectors of zeros and ones, respectively, with their dimensions implied from the context. Sets are denoted by uppercase calligraphy letters, and for any number $k \in \mathbb{Z}_+$, define $[k] := \{1, \dots, k\}$. Given any logical condition "$\cdot$", the function $\mathbb{1}(\cdot)$ is equal to 1 whenever the condition is satisfied, and 0 otherwise.

2. Multi-unit auction design

In this section, we introduce the problem of procurement multi-unit auction design with capacity constraints for minimizing the expected electricity generation cost while satisfying the demand for each time slot (e.g., every hour) of the day. In section Section 2.1 we describe the problem setup and in Section 2.2 the analytical solution for the single time slot setting. In Section 2.3 we reformulate the design problem as a learning problem, extending the learning framework proposed by Dütting et al. (2019) to our setting. In Section 2.4 we present extended formulations that incorporate the key features mentioned in Section 1.

2.1. Multi-unit auction

Consider a multi-unit auction where n generators are competing for $d_j \in \mathbb{R}_{\geq 0}$ identical items, which represent rights to produce one unit of electricity at each time slot $j \in [m]$ in a day. The number of electricity units auctioned is assumed to be equal to a perfectly inelastic aggregate demand d_j to obtain allocations that satisfy the demand requirement.[1] Let $v_i \in \mathbb{R}_{\geq 0}$ be the private unit cost of generator i per unit of electricity across the whole day, such that the additive (linear) cost function for an allocation of y units to this generator is given by $y \cdot v_i$.

Additionally, generator i has constant capacity constraints $\bar{q}_{ij} \in \mathbb{R}_{\geq 0}$ that limit the maximum amount of electricity that it can generate at each time slot j.[2] To ensure competition between generators and circumvent trivial solutions, it is further assumed that the total available capacity generators can supply exceeds the demand $\sum_{i=1}^{n} \bar{q}_{ij} \geq d_j$ at each time slot $j \in [m]$.

Generator i's unit cost v_i is drawn from a marginal distribution F_i. The unit cost profile $\boldsymbol{v} = (v_1, \dots, v_n)$ is unknown to the system operator. However, the underlying joint distribution from which they are drawn $F = (F_1, \dots, F_n)$ is known. Unit costs are assumed to be uncorrelated across generators. Let $\mathcal{V}_i$ be the domain set of all possible values of unit costs v_i of generator i, and denote $\mathcal{V} = \prod_{i=1}^{n} \mathcal{V}_i$, with $\mathcal{V}_{-i} = \prod_{j \neq i} \mathcal{V}_j$.

In an auction, generators are requested to submit their unit cost, which can be misreported, through a single unit-price bid $b_i \in \mathcal{V}_i$. Based on the bidding profiles $\boldsymbol{b} = (b_1, \dots, b_n) \in \mathcal{V}$, the mechanism determines for every time slot the allocation and payments given to the generators.

Definition 2.1 (*Multi-Unit Auction*)**.** Given a bidding profile $\boldsymbol{b} = (b_i)_{i \in [n]} \in \mathcal{V}$, an auction $(\boldsymbol{g}, \boldsymbol{p})$ is characterized by an allocation rule $\boldsymbol{g} = (g_{ij})_{i \in [n], j \in [m]}$ and a payment rule $\boldsymbol{p} = (p_i)_{i \in [n]}$, where $g_{ij} : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ determines the number of units allocated to generator i at time slot j, and $p_i : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ is the payment for the energy produced by generator i under the allocation for the entire day.

Instead of modeling the probability of assigning each unit/item (as in Feng et al. (2018) and Dütting et al. (2019)), here the allocation rule represents the number of units assigned. This is possible in multi-unit auctions since the focus is not on which items, but rather on how many units are assigned. This modification is motivated by three main reasons: (i) under the proposed allocation, the number of parameters in the neural network remains invariant to the number of units being auctioned; (ii) the demand constraint can be easily introduced within the architecture (see Section 3); and (iii) the challenge of determining the metric to express the number of units represented by each right is removed, as the proposed auction directly assigns divisible units to each generator that can be interpreted directly as a fraction of the demand.

Definition 2.2 (*Profit Function with Capacity Constraints*)**.** Each generator with unit cost v_i, under the mechanism $(\boldsymbol{g}, \boldsymbol{p})$, seeks to maximize its daily profit by bidding $b_i \in \mathcal{V}_i$. Given a bidding profile $\boldsymbol{b} \in \mathcal{V}$, profit π_i is defined as the difference between the payment received and the generation cost, if the assigned units are within the capacity for every time slot, and otherwise is considered unbounded $(-\infty)$, i.e.,

$$\pi_i(v_i, \boldsymbol{b}) = \begin{cases} p_i(\boldsymbol{b}) - C(\boldsymbol{g}_i(\boldsymbol{b}), v_i), & \text{if } g_{ij}(\boldsymbol{b}) \leq \bar{q}_{ij} \ \forall j \in [m], \\ -\infty, & \text{otherwise,} \end{cases}$$

where $C(\cdot)$ denotes the cost function, assumed to be positive and linear in both $\boldsymbol{g}_i$ and v_i: $C(\boldsymbol{g}_i(\boldsymbol{b}), v_i) = \sum_{j=1}^{m} g_{ij}(\boldsymbol{b}) v_i$. Note that the subscript i is omitted from C since the dependence on the generator is only given by the inputs.

The optimal design problem considers auctions that follow the *dominant strategy incentive compatibility* (DSIC) and *individual rationality* (IR) conditions, while ensuring that the allocation rule satisfies both the *capacity constraint* (CC) and the *demand constraint* (DC). In the following we define these characteristics and constraints. We denote the unit cost and bidding profiles excluding generator i as $\boldsymbol{v}_{-i} = (v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n)$ and $\boldsymbol{b}_{-i} = (b_1, \dots, b_{i-1}, b_{i+1}, \dots, b_n)$, respectively.

Definition 2.3 (*DSIC*)**.** An auction $(\boldsymbol{g}, \boldsymbol{p})$ is *dominant strategy incentive compatible* (DSIC) if, for every generator, the optimal strategy that maximizes profits is to truthfully report the unit cost, regardless of the other generators' bids,[3] i.e.,

$$\pi_i(v_i, (v_i, \boldsymbol{b}_{-i})) \geq \pi_i(v_i, (b_i, \boldsymbol{b}_{-i})), \quad \forall i \in [n], v_i \in \mathcal{V}_i, b_i \in \mathcal{V}_i, \boldsymbol{b}_{-i} \in \mathcal{V}_{-i}.$$

Definition 2.4 (*IR*)**.** An auction is ex post *individually rational* (IR) if for each generator, bidding truthfully results in a non-negative profit, thus

$$\pi_i(v_i, (v_i, \boldsymbol{b}_{-i})) \geq 0, \quad \forall i \in [n], v_i \in \mathcal{V}_i, \boldsymbol{b}_{-i} \in \mathcal{V}_{-i}.$$

Definition 2.5 (*CC*)**.** An auction satisfies the *capacity constraint* (CC) condition if the allocation rule assigns each generator a quantity that falls within their respective capacity for each time slot, such that

$$g_{ij}(\boldsymbol{b}) \leq \bar{q}_{ij}, \quad \forall i \in [n], j \in [m], \boldsymbol{b} \in \mathcal{V}.$$

Definition 2.6 (*DC*)**.** An auction satisfies the *demand constraint* (DC) if the allocation rule ensures that generators collectively produce enough electricity units to meet or exceed the demand at each time slot, thus

$$\sum_{i=1}^{n} g_{ij}(\boldsymbol{b}) \geq d_j, \quad \forall j \in [m], \boldsymbol{b} \in \mathcal{V}.$$

[1] The assumption of inelastic demand is supported by consumers paying for electricity using fixed tariffs that remain mostly invariant in the short-term.

[2] Since the installed capacity is expected to remain stable, except due to technical reasons, it is considered to be a non-strategic variable that cannot be significantly misreported without attracting considerable attention from the regulator (Balat et al., 2023). Hence, we treat capacities as commonly known parameters.

[3] By initially identifying the optimal strategy for each profit-maximizing generator, and then comparing it with respect to the true unit cost in terms of profitability, the DSIC condition can also be rewritten as the following notion of zero *ex post regret*

$$\max_{v'_i \in \mathcal{V}_i} \pi_i(v_i, (v'_i, \boldsymbol{v}_{-i})) - \pi_i(v_i, (v_i, \boldsymbol{v}_{-i})) = 0, \quad \forall i \in [n], v_i \in \mathcal{V}_i, \boldsymbol{v}_{-i} \in \mathcal{V}_{-i}.$$

Since it is optimal for generators to report truthfully under DSIC mechanisms, the system operator's expected electricity generation cost is defined as

$$cost := \mathbb{E}_{\boldsymbol{v}\sim F}\left[\sum_{i=1}^{n} p_i(\boldsymbol{v})\right].$$

The multi-unit auction design problem aims to find an auction mechanism that minimizes the expected generation cost while satisfying the *incentive compatibility* (DSIC), *individual rationality* (IR), *capacity constraint* (CC) and *demand constraint* (DC) conditions, thus it can be written as

$$\min_{(\boldsymbol{g},\boldsymbol{p})} \quad \mathbb{E}_{\boldsymbol{v}\sim F}\left[\sum_{i=1}^{n} p_i(\boldsymbol{v})\right] \tag{1}$$

$$\text{s.t.} \quad \pi_i(v_i,(v_i,\boldsymbol{b}_{-i})) \geq \pi_i(v_i,(b_i,\boldsymbol{b}_{-i})),$$

$$\forall i\in[n], v_i\in\mathcal{V}_i, b_i\in\mathcal{V}_i, \boldsymbol{b}_{-i}\in\mathcal{V}_{-i}, \tag{DSIC}$$

$$\pi_i(v_i,(v_i,\boldsymbol{b}_{-i})) \geq 0, \quad \forall i\in[n], v_i\in\mathcal{V}_i, \boldsymbol{b}_{-i}\in\mathcal{V}_{-i}, \tag{IR}$$

$$g_{ij}(\boldsymbol{b}) \leq \bar{q}_{ij}, \quad \forall i\in[n], j\in[m], \boldsymbol{b}\in\mathcal{V}, \tag{CC}$$

$$\sum_{i=1}^{n} g_{ij}(\boldsymbol{b}) \geq d_j \quad \forall j\in[m], \boldsymbol{b}\in\mathcal{V}. \tag{DC}$$

2.2. Analytical approach

In settings where there is a single time slot (i.e., $m = 1$), and under some regularity conditions, the optimal mechanism can be solved analytically by following Iyengar and Kumar (2008) and Chaturvedi (2015). Proposition 2.7 characterizes the optimal design for problem (1).

Proposition 2.7 (*Analytical Solution*)**.** *For each generator i, assume that unit costs are independently distributed across generators. In addition, assume that the virtual cost function $v_i + \frac{F_i(v_i)}{f_i(v_i)}$ is non-decreasing in the unit cost v_i (regularity condition). The optimal auction design $(\boldsymbol{g},\boldsymbol{p})$ that solves problem (1) is*

$$g^*(\boldsymbol{v}) = \arg\min_{g} \sum_{i=1}^{n}\left(v_i + \frac{F_i(v_i)}{f_i(v_i)}\right) g_i(\boldsymbol{v}) \quad \textit{s.t.} \quad \sum_{i=1}^{n} g_i(\boldsymbol{v}) \geq d, \quad g_i(\boldsymbol{v}) \leq \bar{q}_i, \ \forall i\in[n],$$

$$p_i^*(\boldsymbol{v}) = v_i g_i^*(\boldsymbol{v}) + \int_{v_i}^{\bar{v}} g_i^*((u_i,\boldsymbol{v}_{-i}))\, du_i, \quad \forall i\in[n],$$

for every unit cost profile $\boldsymbol{v}\in\mathcal{V}$, where the upper limit in the integral $\bar{v}$ corresponds to the highest value in $\mathcal{V}_i$, f_i denotes the density function of v_i. Here the index j is dropped since there is a single time slot.

The proof is available in Appendix C. Observe that if generators have symmetric distributions, the optimal allocation rule assigns full capacity to a subset of the lowest unit cost generators and, at most, its capacity to the *marginal* bidder. In addition, the optimal payment for each generator is discriminatory and covers both the production costs, which ensures participation (IR condition), and the opportunity cost of not revealing their true unit cost, often referred in the literature as the *information rent* (McAfee and McMillan, 1987; McAfee and Reny, 1992).

2.3. Multi-unit auction design as a learning problem

The optimal auction design problem can be reframed as a learning problem, following Dütting et al. (2019). Let $\mathcal{M}$ be the set of auctions that satisfy both the *individual rationality* (IR) condition and the *demand constraint* (DC). The learning problem consists on finding the optimal auction by exploring over a parametric class of auctions $(\boldsymbol{g}^w,\boldsymbol{p}^w)\in\mathcal{M}$ that are encoded by neural networks whose parameters are their weights and biases $\boldsymbol{w}\in\mathbb{R}^r$, with dimension $r>0$.

Apart from the *individual rationality* (IR) condition and the *demand constraint* (DC), which are built into the network architecture as hard constraints (see Section 3), quantifiable and differentiable relaxations are introduced to the other constraints in problem (1). By focusing on constraint satisfaction in expectation and assuming that F has full support over $\mathcal{V}$, deep learning can be leveraged for approximating optimal designs, as initially motivated in Dütting et al. (2015). In an auction $(\boldsymbol{g}^w,\boldsymbol{p}^w)$, relaxing the *dominant strategy incentive compatibility* (DSIC) condition involves ensuring that the expected *ex post* regret, defined as the expected maximum additional profit increase through misreporting, to be 0 for every generator, that is,

$$rgt_i(\boldsymbol{w}) = \mathbb{E}_{\boldsymbol{v}\sim F}\left[\max_{v_i'\in\mathcal{V}_i} \gamma\left(\pi_i^w(v_i,(v_i',\boldsymbol{v}_{-i})) - \pi_i^w(v_i,\boldsymbol{v})\right)\right] = 0, \quad \forall i\in[n].$$

Here $\gamma = \prod_{j=1}^{m} \mathbb{1}\left(g_{ij}^w(v_i',\boldsymbol{v}_{-i}) \leq \bar{q}_{ij}\right)$, i.e., indicator functions are introduced to ensure that the first term remains bounded and capacity constraints are observed when searching for the best possible deviation, in line with Feng et al. (2018).

In addition, penalties are introduced for violating in expectation the *capacity constraint* (CC) for each generator

$$ccp_i(\boldsymbol{w}) = \mathbb{E}_{\boldsymbol{v}\sim F}\left[\sum_{j=1}^{m}\max\left\{g_{ij}^w(\boldsymbol{v}) - \bar{q}_{ij}, 0\right\}\right] = 0, \quad \forall i\in[n].$$

Problem (1) is then formulated as the learning problem

$$\min_{\boldsymbol{w}\in\mathbb{R}^r} \quad \mathbb{E}_{\boldsymbol{v}\sim F}\left[\sum_{i=1}^{n} p_i^w(\boldsymbol{v})\right] \tag{2}$$

$$\text{s.t.} \quad rgt_i(\boldsymbol{w}) = 0, \quad \forall i\in[n]$$

$$ccp_i(\boldsymbol{w}) = 0, \quad \forall i\in[n].$$

To solve (2) we employ a sample of unit cost profiles $\mathcal{S} = \{\boldsymbol{v}^{(\ell)}\}_{\ell\in[L]}$ of size L randomly drawn from distribution F, and compute the expectation terms using empirical averages. The learning problem seeks to minimize the empirical generation cost while ensuring that both the empirical ex post regret and the empirical capacity constraint penalty are zero for all generators, i.e,

$$\min_{\boldsymbol{w}\in\mathbb{R}^r} \quad \frac{1}{L}\sum_{\ell=1}^{L}\sum_{i=1}^{n} p_i^w(\boldsymbol{v}^{(\ell)}) \tag{3a}$$

$$\text{s.t.} \quad \widehat{rgt}_i(\boldsymbol{w}) = 0, \quad \forall i\in[n], \tag{3b}$$

$$\widehat{ccp}_i(\boldsymbol{w}) = 0, \quad \forall i\in[n], \tag{3c}$$

where constraints (3b) and (3c) are defined, respectively, as

$$\widehat{rgt}_i(\boldsymbol{w}) = \frac{1}{L}\sum_{\ell=1}^{L}\left[\max_{v_i'\in\mathcal{V}_i} \hat{\gamma}\left(\pi_i^w(v_i^{(\ell)},(v_i',\boldsymbol{v}_{-i}^{(\ell)})) - \pi_i^w(v_i^{(\ell)},\boldsymbol{v}^{\ell})\right)\right] = 0,$$

$$\widehat{ccp}_i(\boldsymbol{w}) = \frac{1}{L}\sum_{\ell=1}^{L}\left[\sum_{j=1}^{m}\max\left\{g_{ij}^w(\boldsymbol{v}^{(\ell)}) - \bar{q}_{ij}, 0\right\}\right] = 0,$$

$\forall i\in[n]$, with $\hat{\gamma} = \prod_{j=1}^{m}\mathbb{1}\left(g_{ij}^w(v_i',\boldsymbol{v}_{-i}^{(\ell)}) \leq \bar{q}_{ij}\right)$.

The *demand constraint* (DC) and *individual rationality* (IR) conditions are thus met by only considering auctions that satisfy the demand requirement and pay all generators more than their production costs for the allocation.

2.4. Extended formulations

In the remainder of the section, we present variations to the model described above, inspired by the settings presented in Reguant (2014) and Fabra and Llobet (2023), to include other key features of electricity markets mentioned in Section 1.

• *Uncertain capacity and demand.* We first consider the case where available capacity is randomly drawn from a distribution, but each realization is not private information (see[2]). This is motivated by the integration of renewable energy sources into the electricity system, which tend to be less costly, but exhibit greater uncertainty. The analytical solution for this setting is available in Iyengar and Kumar (2008), for the case of a single time slot ($m = 1$) revenue maximizing auction. Next, we consider uncertain demand driven by features that impact the modern electricity consumer, such as on-site generation, electric transport or extreme climate events. To ensure that the auction always meets the *demand constraint* (DC), it is further assumed that, at each time slot, aggregate capacity always covers demand $\sum_{i=1}^{m} \bar{q}_{ij} \geq d_j \; \forall j \in [m]$, for all possible values of $\bar{q}_{ij}$ and d_j, which in this case are random variables.

- *Correlated unit costs.* We consider settings where unit costs are correlated across generators due to their dependence on seasonal variations and fuel prices. This relaxation allows us to determine the impact of correlation on the optimal expected generation cost.

Remark 2.8. The assumption that F has full support over $\mathcal{V}$ might not hold when unit costs are correlated across generators (e.g., for perfectly correlated costs and uniform marginals between 0 and 1, with $\mathcal{V} = [0,1]^2$, the bidding profile $(0,1) \in \mathcal{V}$ has zero probability). When this happens, the learning *in expectation* problem for *revenue maximization* settings results in auctions that avoid allocating items when facing profiles outside the support of the distribution to discourage deviations (refer to Dobzinski et al. (2011) for a detailed discussion). However, for the procurement of essential goods, the threat of not supplying demand lacks credibility. Therefore, problem (3) is designed to optimize over auctions that always procure demand (i.e., for all profiles in $\mathcal{V}$). Nonetheless, this approach brings two additional issues. First, the auction should meet the capacity constraints even for profiles outside the support of the distribution. To achieve this, constraint (3c) is augmented by the capacity constraint violations resulting from profiles where generators misreport optimally, as follows

$$\frac{1}{L}\sum_{\ell=1}^{L}\left[\sum_{j=1}^{m} \max\left\{g_{ij}^{w}(\boldsymbol{v}^{(\ell)}) - \bar{q}_{ij}, 0\right\} + \frac{\beta}{1-\beta}\max\left\{g_{ij}^{w}(v_i'^{*}, \boldsymbol{v}_{-i}^{(\ell)}) - \bar{q}_{ij}, 0\right\}\right] = 0,$$

$\forall i \in [n], \beta \in (0,1)$, where $v_i'^{*} = \arg\max_{v_i' \in \mathcal{V}_i} \hat{\gamma}\pi_i^{w}(v_i^{(\ell)}, (v_i', \boldsymbol{v}_{-i}^{(\ell)}))$. In optimality, with near zero regret, this constraint converges to (3c). A second issue that arises in this case is that the auction may not encourage truthful bidding, irrespective of other generators' reports: when other bidders misreport significantly, pushing the profile beyond the support, the generator can also misreport its unit cost. However, this approach still approximates optimal mechanisms that ensure truthfulness for all profiles within the support of F (commonly referred to as DSIC *in expectation*). To this end, Definition 2.3 is modified to be satisfied $\forall i \in [n], b_i \in \mathcal{V}_i, (v_i, \boldsymbol{b}_{-i}) \in \text{support}(F)$.

Multidimensional bidder types. In the following variation, we consider generators that hold private information across multiple parameters of the profit function. Let $\boldsymbol{v}_i = (v_{ik})_{k\in[h]}$ denote the multidimensional type of generator i with dimension h, denoting the number of parameters over which this private information is defined. Generators can misreport their type by bidding $\boldsymbol{b}_i = (b_{ik})_{k\in[h]}$. Let $\mathcal{V}_i$ and F_i represent the domain set and marginal distribution of generator i's type. Moreover, type and bidding profiles are denoted by $\boldsymbol{v} = (\boldsymbol{v}_i)_{i\in[n]}$ and $\boldsymbol{b} = (\boldsymbol{b}_i)_{i\in[n]}$, respectively. Observe that by substituting v_i and b_i with their vector counterparts $\boldsymbol{v}_i$ and $\boldsymbol{b}_i$, the formulation described in Sections 2.1 and 2.3 can be effectively adjusted to accommodate these modifications. In Section 3, this generalization is already incorporated into the model.

- *Time slot unit costs.* Generators may have different unit costs across different time slots. In fact, several markets allow bids per time slot, usually on an hourly basis (Cramton, 2017; Shah and Chatterjee, 2020). By explicitly considering bids that may vary across time slots we seek to determine whether bundling (i.e., buying electricity units of different time slots as a combined package) or independent auction rules are better for minimizing generation costs. Let $\boldsymbol{v}_i = (v_{ij})_{j\in[m]}$ be the type of generator i, where each v_{ij} represents the unit cost incurred at time slot j and m denotes the number of time slots in a day. In the auction, generators bid m unit-prices $\boldsymbol{b}_i = (b_{ij})_{j\in[m]}$ (one for each time slot). The cost function is thus defined as

$$C(g_i(\boldsymbol{b}), \boldsymbol{v}_i) = \sum_{j=1}^{m} g_{ij}(\boldsymbol{b}) v_{ij}.$$

3. Model architecture

This section describes the neural network architecture and training procedure for computing the allocation and payment rules within the context of procurement multi-unit auctions with n bidders and d_j units to procure at each time slot j. For this, the *RegretNet* framework proposed by Dütting et al. (2019) is adapted to the *fully extended* problem (i.e., the problem incorporating all the variations discussed in Section 2.4).

3.1. Neural network architecture

As illustrated in Fig. 1, the architecture consists of an allocation and a payment component, which are jointly trained as a unified network with parameters $\boldsymbol{w}_g$ and $\boldsymbol{w}_p$, respectively. These components are represented as feed-forward networks that employ a *Leaky ReLU* activation function within their hidden layers, use He initialization (He et al., 2015), and have the bids $\boldsymbol{b}_i$ submitted by each generator as the input layer.

On the one hand, the allocation network encodes the allocation rule to obtain the number of units z_{ij} each generator i will produce at each slot j. After R hidden layers, a *softmax* activation is used to normalize the units within each time slot to be between zero and one, and to sum up to one. Then, these units are multiplied by the demand d_j, resulting in auctions that always procure the aggregate demand $\sum_{i=1}^{n} z_{ij}^{w}(\boldsymbol{b}) = d_j$ at each time slot. On the other hand, the payment network computes the payment each generator receives. Following P hidden layers, the information rent $\tilde{p}_i^{w}$ is estimated. To enforce the information rent to be within $[0,\infty)$, an ELU activation function ($\alpha = 1$) with an addition of $+1$ is employed (Clevert et al., 2015).[4] Subsequently, the generation costs $C(\boldsymbol{z}_{i*}, b_i)$ are added to the information rent, where $\boldsymbol{z}_{i*}$ denotes the allocated units to generator i for each time slot in the day. This ensures that the payment rule effectively covers the costs incurred by each bidder: $p_i^{w}(\boldsymbol{b}) = \tilde{p}_i^{w} + C(\boldsymbol{z}_{i*}, \boldsymbol{b}_i)$, with $C(\boldsymbol{z}_{i*}, b_i) = \sum_{j=1}^{m} z_{ij} b_i$. As a result, the *individual rationality* (IR) condition is satisfied, ensuring that profits always remain non-negative.

Remark 3.1. The proposed payment network differs from that in Bhardwaj et al. (2023), where a multiplier greater than 1 scales the product of bids and allocations. When unit costs approach zero, their model tends to predict zero payments. However, since generators aim to bid higher prices, the optimal auction should incentivize lower-cost generators to bid truthfully with positive payments, especially when other bidders' unit costs are much higher. Our modification achieves this by approximating the information rent, an additional term that ensures individual rationality by making generators' payment exceed

[4] The Exponential Linear Unit (ELU) activation function (Clevert et al., 2015) is defined as follows

$$ELU(x) = \begin{cases} x, & \text{if } x \geq 0 \\ \alpha(\exp(x) - 1), & \text{if } x < 0. \end{cases}$$

The ELU activation function behaves similarly to the $ReLU$ for positive inputs, but it provides a smoother output and prevents the gradient to be zero, which is an important feature when the activation is applied to the output layer.

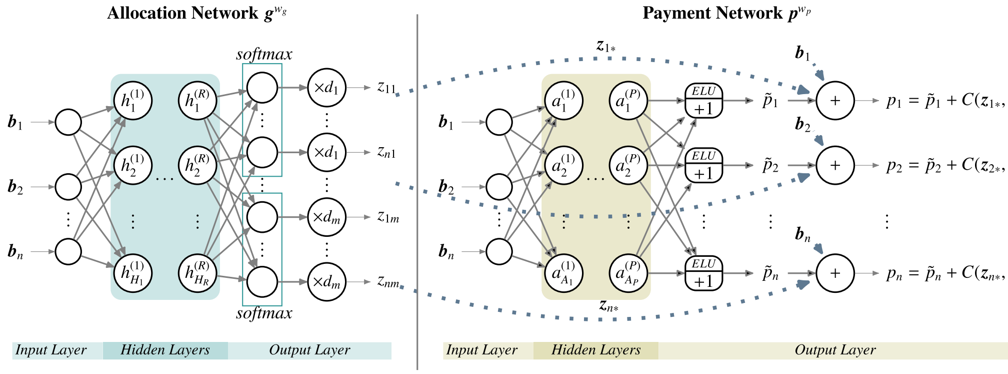


Fig. 1. *RegretNet* architecture for procurement multi-unit auctions, adapted from the multi-item setting of Dütting et al. (2019).

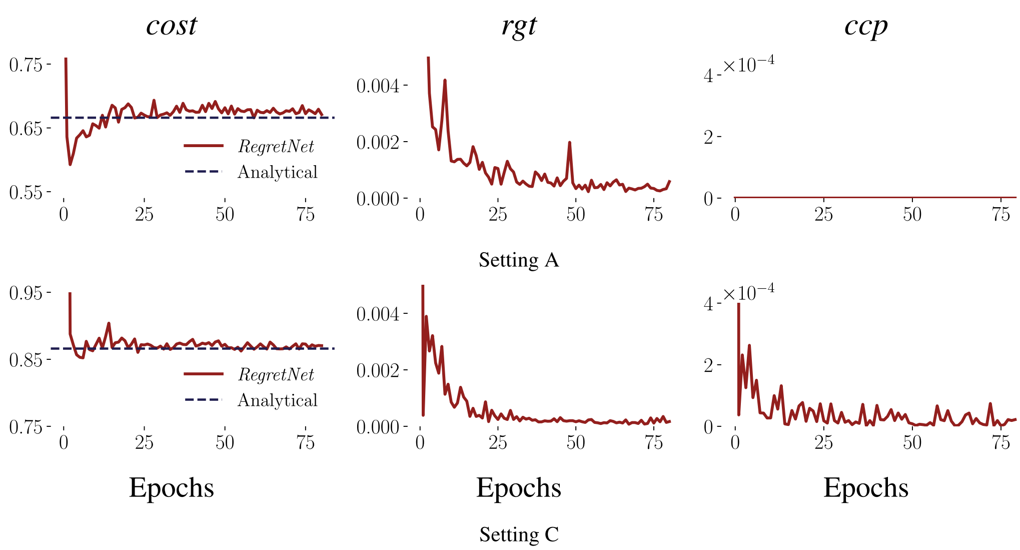


Fig. 2. Learning curves of empirical cost, regret (*rgt*), and capacity constraint penalty (*ccp*), computed during training on the validation set, for settings A and C. The blue dashed lines denote the expected cost of the analytical solution (5). The red curve shows the learning curve for each evaluation metric.

their costs, thereby providing a more general representation of procurement auctions. Also, in optimality, this term will ensure incentive compatibility by properly compensating generators for revealing their true costs (or type).

Remark 3.2. Observe that the proposed network is insensitive to variations in demand scale, as demand is only multiplied by the units in the output layer. The experiments in Section 4 normalize both demand and capacities to be between 0 and 1 for additional stability.

3.2. Training algorithm

The training procedure for learning optimal auctions uses the augmented Lagrangian method to add the constraint violations to the objective function, following Dütting et al. (2019). This method augments the Lagrange formulation with a quadratic penalty term for the violation of each of the constraints.[5] Problem (3) turns into minimizing the following unconstrained loss function

$$\mathcal{L}_\rho(\boldsymbol{w},\lambda) = \frac{1}{L}\sum_{\ell=1}^{L}\sum_{i=1}^{n} p_i^w(\boldsymbol{v}^{(\ell)}) + \sum_{i=1}^{n} \lambda_i^{rgt}\widehat{rgt}_i(\boldsymbol{w}) + \frac{\rho^{rgt}}{2}\sum_{i=1}^{n}\left(\widehat{rgt}_i(\boldsymbol{w})\right)^2 + \sum_{i=1}^{n} \lambda_i^{ccp}\widehat{ccp}_i(\boldsymbol{w}) + \frac{\rho^{ccp}}{2}\sum_{i=1}^{n}\left(\widehat{ccp}_i(\boldsymbol{w})\right)^2, \tag{4}$$

where $\lambda^{rgt} \in \mathbb{R}^n$ and $\lambda^{ccp} \in \mathbb{R}^n$ represent the vectors of the Lagrange multipliers, and $\rho^{rgt} > 0$ and $\rho^{ccp} > 0$ denote the weights of the squared penalties for the violations associated with the two sets of constraints: the zero empirical ex post regret constraint, and the capacity constraints. The training procedure has the following schedule for updating the parameters:

(1) In each iteration, update $\boldsymbol{w}_{t+1} \leftarrow \arg\min_{\boldsymbol{w}} \mathcal{L}_{\rho_t}(\boldsymbol{w}_t, \lambda_t)$.
(2) Every Q_λ^{rgt} iterations, update $\lambda_{i,t+1}^{rgt} \leftarrow \lambda_{i,t}^{rgt} + \rho_t^{rgt}\widehat{rgt}_i(\boldsymbol{w}_{t+1})$ for each generator.
(3) Every Q_λ^{ccp} iterations, update $\lambda_{i,t+1}^{ccp} \leftarrow \lambda_{i,t}^{ccp} + \rho_t^{ccp}\widehat{ccp}_i(\boldsymbol{w}_{t+1})$ for each generator.

The weights $\rho^{rgt} > 0$ and $\rho^{ccp} > 0$ are incremented by the given constants $\varepsilon^{rgt} > 0$ and $\varepsilon^{ccp} > 0$, every Q_p^{rgt} and Q_p^{ccp} iterations, respectively. Moreover, at the beginning of each iteration, an inner optimization loop is solved to compute the empirical regret. This loop finds the optimal deviation (misreport) from the true unit cost, by minimizing the negative utility for each generator, using multiple

[5] For context, when only penalizations are included, achieving near-zero constraint violations requires increasing the parameters ρ in (4) to infinity. The Augmented Lagrangian Method, through the estimation of the Lagrange multipliers, circumvents the need for $\rho \to \infty$ to attain convergence towards the optimal solution, thus avoiding ill-conditioning.

random initializations. Additional details for the training procedure are given in Appendix A.

4. Experiments

In this section, we present a series of experiments to evaluate the performance of the proposed method. We employ both (i) synthetic datasets, generated from common choices for distribution F in the auction design and electricity generation literature; and (ii) real electricity market data, from which the underlying distributions are estimated for profile sampling. We start with settings involving a single time slot ($m = 1$) and independently drawn unit costs from the uniform distribution (Section 4.1.1).[6] Settings with correlated unit costs across generators follow in Section 4.1.2. Next, we relax the assumption about constant capacities and demand, exploring cases with noise, supply failure risk, and wind and solar power integration (Section 4.1.3). Experiments with multiple time slots $m > 1$ in a day follow, including scenarios where generators have either daily unit costs (Section 4.2.1) or different unit costs for each time slot (Section 4.2.2). Finally, in Section 4.3, we conduct experiments based on the Colombian electricity market as a case study.

Setup. The learning algorithm for each setting is implemented in Tensorflow 2.12 and run in a compute cluster. In all experiments, datasets were created by drawing samples of profiles from the chosen (synthetic datasets) or estimated (real datasets) distributions. We used mini-batch sizes of $\{128, 256, 350\}$ profiles and a number of batches in $\{100, 200, 300\}$ for training, and 50 batches for validation. The inner optimization was initialized with two misreports and 30 steps were conducted for each one. The algorithm was run for a maximum of 80 epochs. For the testing phase, the empirical cost was computed using a sample of 250,000 profiles for robustness, and the empirical constraint violations were evaluated in 5120 profiles. For the inner optimization, 100 misreports were used, and 2000 gradient-descent steps were performed for each misreport. Additional hyper-parameters are shown in Table A1.

While in supervised learning, model performance is easily assessed by comparing predictions to the observed targets in the training dataset, for this specific learning problem, monitoring the algorithm progress during training and evaluating its optimality is more complex. To address this, we compute the empirical cost and constraint violations for the validation set at each epoch to check convergence stability. Additionally, the results are evaluated on a separate test set, and for robustness, we compare the solution with the analytical solutions, when available.

4.1. Single time slot auctions

In this section, we start by assuming that unit costs are uncorrelated across generators in Section 4.1.1, and then in Section 4.1.2 we study settings with correlated costs. In Section 4.1.3 we relax the assumption of constant capacity and demand, treating them as random variables.

4.1.1. Uncorrelated unit costs

We begin with experiments where unit costs are uncorrelated across generators, for which analytical solutions are presented in Proposition 2.7. These experiments aim to demonstrate that solving the learning problem (3) can recover known analytical solutions. We further assume that generators are competing to meet a unitary demand ($d = 1$) and unit costs are drawn from the distribution $U[0, 1]$, considering the following settings:

A 2 generators with no capacity constraints ($\bar{q}_1, \bar{q}_2 \geq 1$).
B 2 generators with the same capacity of 0.6 each.
C 2 generators: the first with a capacity of 0.6 and the second of 0.8.
D 2 generators: the first with a capacity of 0.3 and the second of 0.9.
E 3 generators: the first with a capacity of 0.6, while the second and third have a capacity of 0.4 each.

From Proposition 2.7 (which builds upon (Iyengar and Kumar, 2008) and Chaturvedi (2015)) we derive the following closed-form analytical solution for the previous settings with 2 generators

$$g_i^*(\boldsymbol{v}) = \begin{cases} \min(d, \bar{q}_i), & \text{if } v_i < v_k, \\ d - \min(d, \bar{q}_k), & \text{if } v_i > v_k, \end{cases}$$
$$p_i^*(\boldsymbol{v}) = \begin{cases} v_i g_i^*(\boldsymbol{v}) + (v_k - v_i) g_i^*(\boldsymbol{v}) + (1 - v_k)(d - \min(d, \bar{q}_k)), & \text{if } v_i < v_k, \\ v_i g_i^*(\boldsymbol{v}) + (1 - v_i) g_i^*(\boldsymbol{v}), & \text{if } v_i > v_k, \end{cases} \tag{5}$$

for each generator $i \in \{1, 2\}$, with $i \neq k$, and for every $\boldsymbol{v} \in [0, 1]^2$ with $d = 1$. If $v_i = v_k$ the allocation and payments are uniformly randomized between the two cases shown in (5). Note once more that the optimal payment for each generator covers both the cost incurred in generating the allocated units and the opportunity cost associated with increasing the unit-price bid.

Fig. 2 displays, for settings A and C, the learning curves of the empirical cost across training epochs achieved in the validation set. In addition, we evaluate the average regret (rgt) and capacity constraint penalty (ccp) across bidders and profiles, i.e., $rgt = \frac{1}{n} \sum_{i=1}^{n} \widehat{rgt}_i(\boldsymbol{w})$ and $ccp = \frac{1}{n} \sum_{i=1}^{n} \widehat{ccp}_i(\boldsymbol{w})$. The extended *RegretNet* converges relatively fast towards the expected cost of the optimal mechanism (i.e., of the analytical solution) and achieves low violations of the regret and capacity constraints. Table 1 summarizes the performance of the auction learned on the test set and compares it with the known analytical solution. Across all settings, the proposed method successfully recovers optimal cost levels (with errors $< 1\%$), incurring in low constraint violations (≤ 0.001).

The last two columns of Table 1 show the expected costs of the pay-as-bid and uniform-price auctions for settings with equilibrium solutions in Chaturvedi (2015) and Fabra and Llobet (2023).[7] In the symmetric settings (A and B), the optimal auction shares the same expected generation cost with the pay-as-bid and uniform auctions. In Fig. 3, we plot the equilibrium bid functions under each auction. While in setting A (uncapacitated generators), the uniform-price and the optimal auctions are identical, with capacitated generators (setting B), both the pay-as-bid and uniform-price auctions fail to satisfy *incentive compatibility*, as generators bid well above their unit costs. As the capacity asymmetry grows, the payments in the pay-as-bid auction increasingly exceed those in the optimal auction (Table 1).[8] Although in setting C the learned auction does not outperform the pay-as-bid due to minor prediction errors, it achieves a 1.42% cost improvement in setting D, where significant capacity asymmetries exist.

Fig. 4 provides a visual representation of the allocation and payment rules of both the analytical solution (left block) and the corresponding

[6] The uniform distribution is a common starting point in the auction design literature, as it helps to illustrate and analyze the optimal auction properties before addressing more complex cases.

[7] The pay-as-bid auction, also known as the discriminatory auction, sequentially allocates units to generators with the lowest offered prices, exhausting their capacity or until demand is satisfied, and pays winners their submitted unit price. In a uniform-price auction, all winners are instead paid a unit price equal to the bid of the marginal (last winning) generator.

[8] In symmetric settings, the pay-as-bid, uniform-price and optimal auctions have the same expected cost, due to the Revenue Equivalence Theorem, which states that mechanisms leading to identical outcomes have the same expected revenue (Klemperer, 1999) In procurement multi-unit auctions, mechanisms that allocate units to the lowest-cost bidders will have equal expected costs. This does not hold in the pay-as-bid auction in asymmetric cases, as the lowest-cost supplier may not submit the lowest bid (Chaturvedi, 2015).

Table 1
Summary of the performance in the test set in terms of cost, regret, and capacity constraint penalty for settings A–E. The expected cost for the pay-as-bid and uniform-price auctions were computed using the equilibrium characterizations of Chaturvedi (2015) and Fabra and Llobet (2023).

		Optimal	*RegretNet*			Pay-as-bid	Uniform-price
		cost	*cost*	*rgt*	*ccp*	*cost*	*cost*
A	*Uncapacitated*	0.6664	0.6691	<0.001	–	0.6664	0.6664
B	$\bar{q} = (0.6, 0.6)$	0.9333	0.9318	0.001	<0.001	0.9333	0.9333
C	$\bar{q} = (0.6, 0.8)$	0.8665	0.8693	<0.001	<0.001	0.8690	
D	$\bar{q} = (0.3, 0.9)$	0.9333	0.9350	<0.001	<0.001	0.9485	
E	$\bar{q} = (0.6, 0.4, 0.4)$	0.8002	0.7978	<0.001	<0.001		

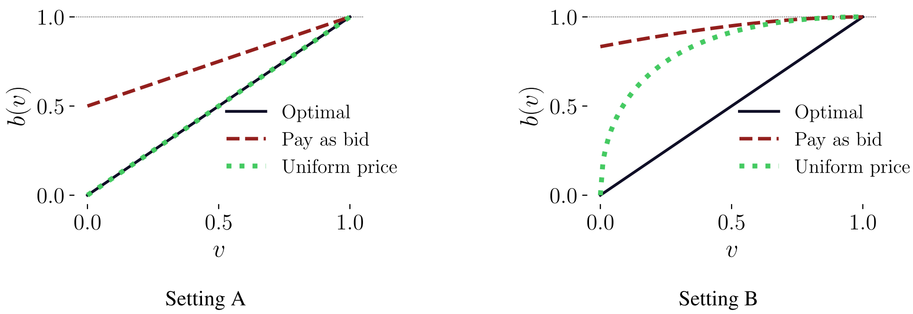


Fig. 3. Equilibrium price-unit bids against unit cost levels for symmetric settings (A and B). The bid function for the optimal mechanism (5) is denoted by the blue line, for the pay-as-bid auction (Chaturvedi, 2015; Fabra and Llobet, 2023) by the red-dashed line, and for the uniform-price auction (Fabra and Llobet, 2023) by the green-dotted line.

RegretNet approximation (right block) for settings A and C (for settings B and D, refer to Figure B4 in the Appendix). The rules $(g_i(\boldsymbol{v}), p_i(\boldsymbol{v}))$ for each generator i are presented as functions of the unit costs (v_1 and v_2), with contour levels denoting the number of units assigned (in the allocation rule) and the corresponding payment (in the payment rule). The learned rule is smoother at the discontinuities as expected, but the approximation of both the allocation and payment rules is overall accurate. Observe that the auctions sequentially assign units to generators with the lowest unit costs, exhausting their capacity or until demand is satisfied, and pay generators an amount that cannot be increased by misreporting unilaterally. For instance, in setting C, above the dotted black line, when $v_1 < v_2$, the auction assigns 0.6 units to the first generator, which corresponds to its capacity, and then allocates the remaining 0.4 units to the second generator to meet the demand. In addition, the first generator receives the same or a lower payment if it bids a higher value at any unit cost profile.

During the analysis of the solution learned in the previous settings, some of the key insights of the optimal mechanism are identified. First, the design achieves productive efficiency as it allocates more units to the lowest-cost generators. Second, for a particular generator, the higher the unit costs of the other bidders are, the higher the payment for that generator. This can be attributed to the increased opportunity cost of misreporting, which therefore requires an increase in the information rent to ensure truthfulness. Third, an increase in aggregate capacity across generators leads to a decrease in the expected cost. This occurs since the allocation rule is now allowed to assign more generation units to lower-cost generators. Finally, distributing aggregate capacity among more generators reduces the expected cost due to increased competition.

Finally, in the optimal auction, payments increase with capacities and always cover the incurred costs. Thus, even with private capacity, generators do not have incentives to withhold it. Moreover, overbidding capacity leads to negative and unbounded profits (Definition 2.2). Iyengar and Kumar (2008) prove that when both costs and capacities are privately held, it is optimal not to offer any information rent for truthfully revealing the capacity. Therefore, the optimal allocation and payment rules remain consistent with those in settings where capacity is known (as shown in Appendix B.1). Nonetheless, expected costs vary depending on the distribution of capacities as bidder strategies are influenced by capacity levels.

> **Takeaway 1.**
>
> I. *The proposed method recovers the optimal auction outlined in Proposition 2.7, attaining low approximation errors in cost levels (<1%) and constraint violations (≤0.001).*
> II. *Increasing aggregate capacity or distributing it among more generators leads to lower expected generation costs.*
> III. *Settings with private and known capacities result in the same expected costs, as generators have no incentives to misreport capacities.*

4.1.2. Correlated unit costs

We now consider settings where unit costs are correlated across generators, driven by factors such as geographical proximity, shared operating conditions, and similar fuel sources (e.g., in thermal generation). In these experiments, two generators compete for a unit demand and the joint distribution of profiles $F = (F_1, F_2)$ is modeled using a Gaussian Copula to introduce dependence, with σ denoting the level of correlation, and each marginal following a uniform $U[0, 1]$ distribution.[9] We consider the following scenarios with two generators:

- F No capacity constraints $\bar{q}_1, \bar{q}_2 \geq 1$. Perfectly positively correlated unit costs ($\sigma = 1$).
- G Capacities of 0.6 and 0.8. Perfectly positively correlated unit costs ($\sigma = 1$).
- H Capacities of 0.6 and 0.8. Positively correlated unit costs ($\sigma = 0.5$).
- I No capacity constraints $\bar{q}_1, \bar{q}_2 \geq 1$. Perfectly negatively correlated unit costs ($\sigma = -1$).

[9] We employ copulas to model dependence between bidder types (see Appendix A.2).

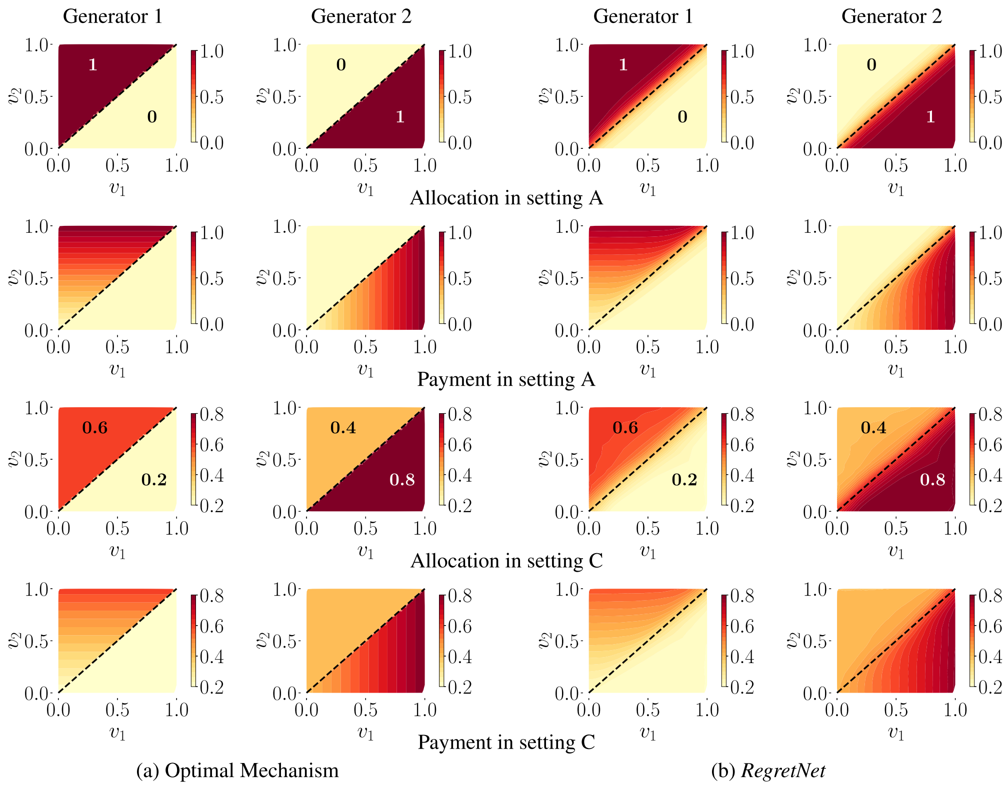


Fig. 4. Comparison of the allocation rule between the optimal mechanism (5) and the auction learned. Note that the color bar ranges are different across settings. The left-most block illustrates the analytical solution, and the right-most block shows the learned rule for each generator. Dashed lines are used to separate the regions for each case (in (5)). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Summary of the performance in the test set in terms of cost, regret, and capacity constraint penalty for settings F–K. The expected cost in mechanism (5) is shown for settings with correlation in unit costs.

			Mech. (5)	*RegretNet*		
			cost	*cost*	*rgt*	*ccp*
F	*Uncapacitated*	$\sigma = 1$	0.5000	0.4954	<0.001	–
G	$\bar{q} = (0.6, 0.8)$	$\sigma = 1$	0.8021	0.8051	0.001	<0.001
H	$\bar{q} = (0.6, 0.8)$	$\sigma = 0.5$	0.8440	0.8408	0.001	<0.001
I	*Uncapacitated*	$\sigma = -1$	0.7497	0.5200	<0.001	–
J	$\bar{q} = (0.6, 0.8)$	$\sigma = -1$	0.8999	0.7693	0.001	<0.001
K	$\bar{q} = (0.6, 0.8)$	$\sigma = -0.5$	0.8838	0.8593	0.001	<0.001

J Capacities of 0.6 and 0.8. Perfectly negatively correlated unit costs ($\sigma = -1$).

K Capacities of 0.6 and 0.8. Negatively correlated unit costs ($\sigma = -0.5$).

Fig. 5 illustrates the learning curves for the cost in experiments F–K in the validation set. The green dash-dotted line denotes the expected cost from mechanism (5). In positively correlated settings F–H, the learned auction cost converges to the expected cost of mechanism (5), while in negatively correlated experiments I–K, auctions tend to achieve lower costs. As shown in Table 2, all learned auctions have low constraint violations (≤0.001), with positively correlated settings showing costs within 1% of the expected cost in mechanism (5). All settings with correlation exhibit lower expected costs when compared to the uncorrelated cases A and C. Moreover, with capacity-constrained generators, positively correlated unit costs consistently result in lower expected costs than their negatively correlated counterparts.

Positive correlation. In the optimal design literature, the mechanism proposed by Crémer and McLean (1988) can extract full surplus even from slightly positively correlated types. By focusing on *interim individual rationality* (IR) (on expectation over types of other bidders), rather than *ex post* IR, the mechanism can pay bidders less than their costs through participation fees. However, it is often considered non-credible, as bidders might not participate in hindsight (Segal, 2003). Table 2 shows that *ex post* IR designs, i.e., *á la* Myerson mechanisms, such as the one in Proposition 2.7, remain optimal in settings with correlation, which is consistent with Roughgarden and Talgam-Cohen (2016), where this result is proved for symmetrical settings. Moreover, our findings show that this result holds even when generators have asymmetrical capacity constraints.[10]

In experiment F, with perfectly positive correlation and without capacity constraints, the optimal auction achieves full surplus extraction: the expected cost is equal to the expected value of the uniform distribution between 0 and 1, which is the unit cost distribution. An auction that pays exactly the production costs to each generator induces competition for maximizing the allocated units and removes incentives to misreport. This aligns with the observation of McAfee and Reny (1992), where identical bidder types lead to a Bertrand-like competition where private information is revealed, and bidders have zero profits. As shown in Fig. 6 (top row), which displays the learned allocation and payment rules for each generator, the auction pays generators only their costs, providing no information rent. For instance, for any profile

[10] These results also indicate that in positively correlated settings, the DSIC in expectation mechanism (the learned auction) can be implemented as an optimal DSIC mechanism (i.e., the analytical solution of Proposition 2.7) as they both result in the same expected costs.

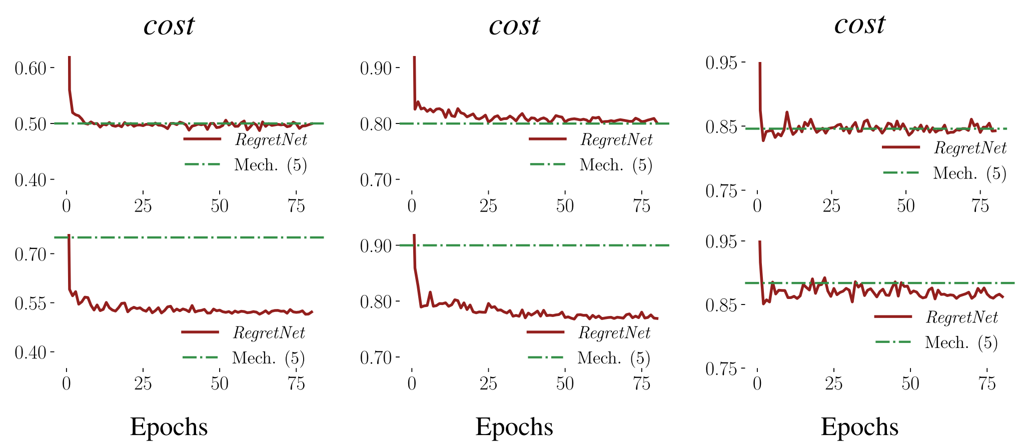


Fig. 5. Learning curves of empirical cost during training in the validation set. The top row shows settings F–H, while the bottom row, I–K, arranged from left to right. The green dashed–dotted lines denote the expected cost of mechanism (5) with correlated unit cost profiles.

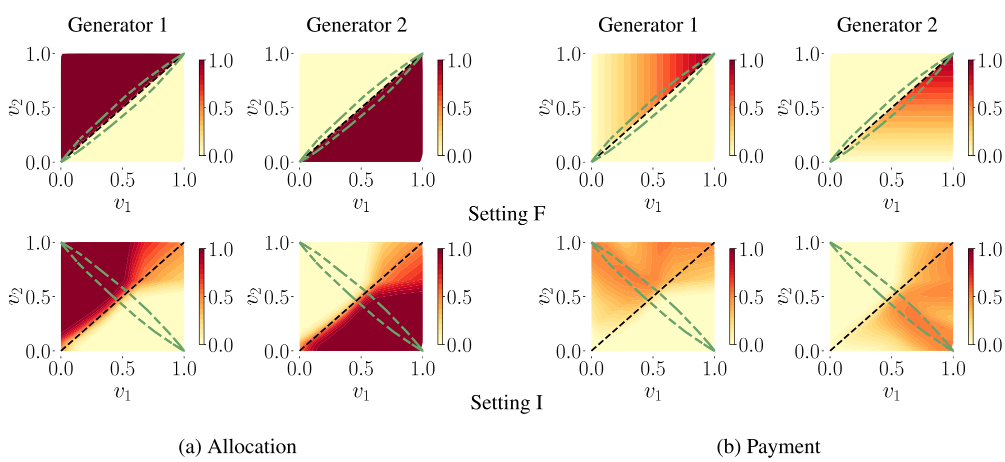


Fig. 6. Auction rules learned in settings F and I for each generator. The left-most block shows the allocation rule, and the right-most block the payment rule. Black dashed-lines are used to separate the decision boundaries. The entire domain of profiles is shown, but the support of the joint distribution lies only within the green-dashed ellipse.

within the support (inside the green ellipse), when $v_1 < v_2$, generator 1 is paid only its unit cost, and if it rises its bid unilaterally, it is assigned zero units.

In Fig. 7, the average information rents of experiments H (red line, $\sigma = 0.5$) and C (green line, $\sigma = 0$) are shown against the generator 2's unit costs v_2, discretized in intervals. Setting H results in a lower expected cost, mainly because profiles with opposite unit costs across bidders are less common, resulting in more price competition and smaller information rents overall. Although these findings indicate that diversification strategies aimed at decreasing the correlation between unit costs may increase average generation costs, in practice, these strategies are more directed to improve operational reliability and, in any case, generators often hedge against the risk of high unit costs to increase profit stability.

Negative correlation. We now consider scenarios with negative correlation, i.e., experiments I–K, for which analytical results are limited.[11] In setting I without capacity constraints, the auction rules are shown in Fig. 6 (bottom row). Observe that the learned mechanism increased in complexity as it leverages the negative correlation to pay lower information rents while achieving incentive compatibility.

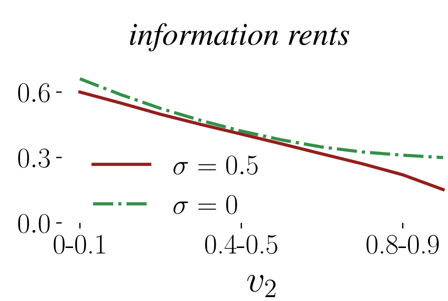


Fig. 7. Average information rents (generation costs minus costs incurred by generators) plotted against the unit costs of the second generator, discretized in intervals of 0.1, for settings H (red curve) and C (green dashed–dotted curve). Information rents are averaged within the intervals and across the unit costs of the first generator.

By first ensuring truthful bidding in the highest-cost generator, the auction infers the approximate value of the unit cost of the lower-cost generator. All units are assigned to the lowest-cost generator if its bid is below 0.5, otherwise, assigned units decrease progressively. The lowest-cost generator is paid 0.5 if it truthfully reports a cost near 1 minus the highest unit cost. If this generator bids untruthfully, but below 0.5, it is paid its bid per unit. For bids exceeding 0.5, each generator is paid the other bidder's unit cost. Thus, the maximum

[11] In electricity auctions, there might be complementarities between different generation sources (e.g., between gas and fuel oil or hydro and solar power) which might result in some level of negative correlation between unit costs.

possible payment for the lowest-cost generator through misreporting is about 0.5, matching the payment when bidding truthfully.

In experiment J the optimal auction also prevents misreporting by leveraging the negative correlation between unit costs (Figure B6 in the Appendix). However, when generators are capacitated, they face the tradeoff of increasing the bid while forgoing some units, leading to higher information rents overall: the generator with the largest capacity faces a higher opportunity cost of misreporting, as it secures a significant portion of units in the allocation, even if it misreports. A similar pattern is found in setting K of imperfect negative correlation, where the learned auction can reduce the expected cost of the design (5), albeit less effectively than in the perfect correlation case (as illustrated in Figure B6 in the Appendix).

Takeaway 2.

I. *The optimal auction when unit costs are positively correlated remains the same as in uncorrelated settings, but leads to lower expected costs due to a higher probability of instances where unit costs are similar between generators, reducing information rents overall. This holds even in scenarios with asymmetrical capacity constraints.*

II. *The optimal auction when unit costs are negatively correlated is different and leads to lower expected costs, compared to uncorrelated settings, as it can leverage this correlation to approximate the unit cost of the lower-cost generator, reducing information rents.*

4.1.3. Uncertain capacity and demand

Integrating renewables has been shown to reduce energy prices, but it can increase supply uncertainty (Sensfuß et al., 2008; Brouwer et al., 2014). At the same time, rising on-site generation, electric transport, and extreme climate events have raised demand variability. Frequent instances of high demand coupled with low supply capacity may lead to greater reliance on more expensive energy units. Motivated by these considerations, the following experiments estimate how uncertainty impacts generation costs in optimal auctions. Here we assume that capacity and demand are random variables whose realizations are common knowledge, i.e., they are non-strategic and not regarded as private information (see[2]). Moreover, unit costs and capacities are independently drawn from known distributions.

Two generators, one with normally-distributed capacity.

L 2 generators: the first one with capacity drawn from a truncated normal distribution centered at 0.6, a standard deviation of 0.3, and ranging from 0.2 to 1. The second generator has a constant capacity of 0.8. Unit costs are drawn from $U[0,1]$ and demand is 1.

This setting includes a generator with an uncertain capacity centered at 0.6. When compared to experiment C, where generator 1 has a constant capacity of 0.6, both auctions result in similar expected costs (Table 3). In Fig. 8, violin plots show the distribution of the generation cost for different values of the capacity levels of generator 1 ($\bar{q}_1$), with unit cost profiles sampled randomly. In addition, this figure includes a histogram of $\bar{q}_1$ at the top and generation costs on the right in black, with green bars representing the generation cost histogram for C. Note that when capacity decreases, costs increase on average compared to when capacity is always 0.6. This is due to the need to resort to other generators to meet demand when the first bidder has low capacity, even if these additional units have higher unit costs. However, this is offset by instances where this bidder has higher capacity and lower unit cost levels, allowing the auction to access less expensive units. In addition, the generation cost is more variable, compared to the cost when the capacity is always 0.6.

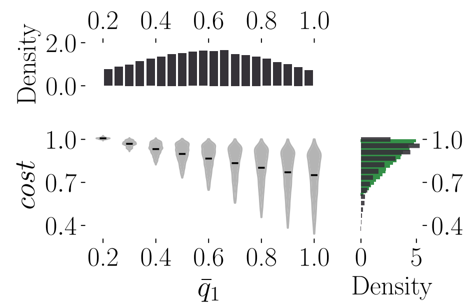


Fig. 8. Violin plots display the distribution of generation costs for multiple fixed values of capacity of the first generator for setting L, using a sample of 10,000 unit cost profiles at each value. On top the histogram of the capacity of the first generator is displayed, while on the right the black bars depict a histogram of the cost. Also on the right, the cost of the optimal auction for setting C is depicted in green. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

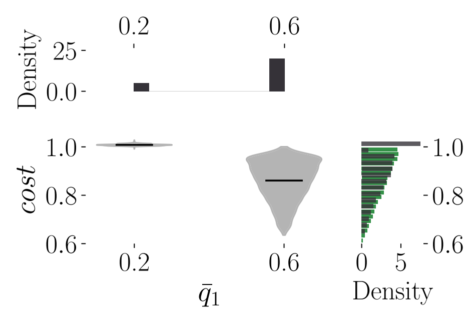


Fig. 9. Violin plots display the distribution of generation costs for multiple values of capacity of the first generator for setting M. The histogram of the cost of the optimal auction for setting C is in green. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Two generators, one with bimodal capacity.

M 2 generators: the first one with capacity of 0.6 with an 80% probability, and capacity of 0.2 with 20% probability. The second generator has a constant capacity of 0.8. Unit costs are drawn from $U[0,1]$ and demand is 1.

This experiment examines variable power generation reliability, which can lead to supply failure risk. In Fig. 9, observe that in instances of low capacity ($\bar{q}_1 = 0.2$), costs escalate considerably. Since there is no reserve capacity, when the first generator fails to deliver, the operator has to resort to potentially higher unit cost generators to satisfy the demand, driving up generation costs. As a result, compared to setting C, where capacity is always 0.6, in setting M the auction incurs in higher expected costs (see also Table 3).

Fixed capacity and normally-distributed demand.

N 2 generators: the first with a capacity of 0.6 and the second with a capacity of 0.8. Unit costs are drawn from $U[0,1]$ and demand from a truncated normal distribution centered at 1, with standard deviation of 0.3, and ranging between 0.6 and 1.4.

In this experiment demand is uncertain and follows a truncated normal distribution. Fig. 10 illustrates a significant rise in costs as demand increases, particularly above the mean of 1. Expected costs are greater compared to setting C, where demand is always 1 (Table 3). This difference arises from the operator purchasing more units from costly generators and the increased opportunity cost of misreporting for the lowest-cost generator during periods of high demand. While in settings

Table 3
Summary of the performance in the test set for settings L–P. In setting P, the bimodal distribution using Beta unimodals is described as $xB_1 + (1-x)B_2$, where $B_1 \sim \text{Beta}(2, 3.5)$, $B_2 \sim \text{Beta}(0.8, 7.2)$, and $x \sim \text{Bernoulli}(0.8)$.

	$\bar{q}$	d	*RegretNet*		
			cost	*rgt*	*ccp*
	$\boldsymbol{v_i \sim U[0,1]}$				
L	$\bar{q}_1 \sim \text{TN}(0.6, 0.3, 0.2, 1)$; $\bar{q}_2 = 0.8$	1	0.8674	<0.001	<0.001
M	$\bar{q}_1 = 0.4x + 0.2, x \sim \text{Bern}(0.8)$; $\bar{q}_2 = 0.8$	1	0.8907	<0.001	<0.001
N	$\bar{q}_1 = 0.6$; $\bar{q}_2 = 0.8$	$\text{TN}(1, 0.3, 0.6, 1.4)$	0.8849	<0.001	<0.001
	$\boldsymbol{v_1 \sim U[0, 0.4]; v_2, v_3 \sim U[0,1]}$				
O	$\bar{q}_1 \sim \text{Rayleigh}(0.3)$; $\bar{q}_2, \bar{q}_3 = 0.5$	1	0.6770	<0.001	<0.001
P	$\bar{q}_1 \sim$ Bimodal Beta; $\bar{q}_2, \bar{q}_3 = 0.5$	1	0.7336	<0.001	<0.001

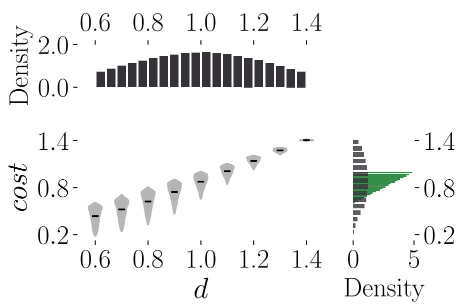


Fig. 10. Violin plots display the distribution of generation costs for multiple values of demand for setting N. The histogram of the cost of the optimal auction for setting C is in green. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

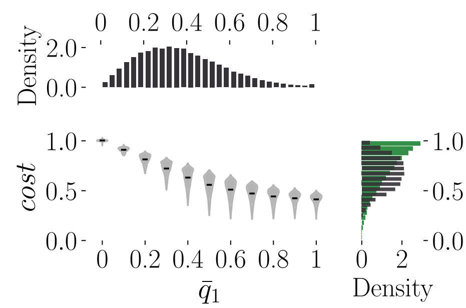


Fig. 11. Violin plots display the distribution of generation costs for multiple values of capacity of the first generator for setting O. The green histogram corresponds to a setting with three bidders, each having a constant capacity of 0.5 and unit costs drawn from $U[0,1]$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

M and N, the expected cost only increases marginally compared to C, when capacity is low or demand is high in these settings, operational reliability and price stability become a concern.

Three generators, one with Rayleigh capacity (wind), two fixed.

O 3 generators: the first with capacity drawn from a Rayleigh distribution with a scale parameter of 0.3 and with unit costs drawn from $U[0, 0.4]$. The second and third generators have a constant capacity of 0.5 units each, and their unit costs are drawn from $U[0,1]$. Demand is unitary.

In this setting we incorporate a wind generator with capacity distributed according to the Rayleigh distribution, which is often used to model wind speed (Manwell et al., 2010; Mazidi et al., 2014). For simplicity, the generated output power corresponds directly to the wind speed, ignoring minor non-linearities and critical speed thresholds. Compared to the optimal auction in scenarios with 2 or 3 generators with constant capacities of 0.5 each, and unit costs drawn from $U[0,1]$ (expected costs of 1 and 0.7508, respectively), the approximated auction has lower expected costs, representing reductions of 32.3% and 9.8% each (Table 3). In Fig. 11 we observe the decline in generation costs as the capacity of the wind generator increases. This reduction occurs because the auction supplements the generation mix with a generator that has on average lower costs per unit, often referred to as the merit-order effect, as depicted in Fig. 13.

Three generators, one with bimodal-beta capacity (solar), two fixed.

P 3 generators: the first with capacity drawn from a bimodal distribution: 80% from a Beta distribution with parameters $\alpha = 2, \beta = 3.5$, and 20% from a Beta distribution with $\alpha = 0.8, \beta = 7.2$. Unit costs are drawn from $U[0, 0.4]$. The second and third generators have a constant capacity of 0.5 units each and their unit costs are drawn from $U[0,1]$. Demand is unitary.

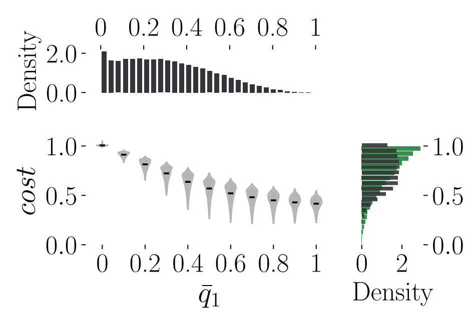


Fig. 12. Violin plots display the distribution of generation costs for multiple values of capacity of the first generator for setting P. The green histogram corresponds to a setting with three bidders, each having a constant capacity of 0.5 and unit costs drawn from $U[0,1]$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

This setting includes a solar generator with capacity distributed according to a bimodal distribution, where a Beta distribution is used for each unimodal. This distribution is commonly used to model solar irradiance (Atwa et al., 2009; Mazidi et al., 2014). Output is assumed to follow the same distribution as solar irradiance. The approximated auction has a lower expected cost, with a reduction of 26.6% and 2.3% compared to scenarios with 2 or 3 generators with capacities of 0.5 each, and unit costs drawn from $U[0,1]$, respectively (Table 3). Fig. 12 shows a decrease in generation costs as the solar generator's capacity increases. This decline is again attributed to the operator's access to lower-cost units, thereby supplementing the generation mix with renewables, as shown in Fig. 13.

These results show some of the challenges of integrating renewables into the dispatch problem. As renewable capacity decreases, generation

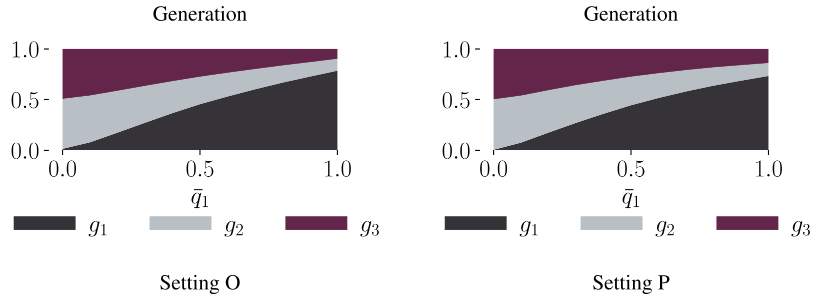


Fig. 13. Generation mix as a function of the first generator capacity for settings O–P. The generation mix shows the participation of each generator in the allocation.

costs — and consequently, electricity prices — can increase significantly, raising concerns for operational reliability and price stability. In addition, supply failures are more prevalent with renewables due to the complexity of accurately forecasting wind or sunlight availability, further intensifying the issue. Therefore, an interesting open question for future work is to explore the effect of reserve capacity or demand programs to alleviate price fluctuations in these instances.

Takeaway 3.

I. Settings with supply failure risk and uncertain demand lead to higher and more variable generation costs.

II. Integrating wind and solar generators decreases expected costs and supplements the generation mix, compared to settings with generators with constant capacities and higher unit costs.

4.2. Multiple time slots auctions

Experiments are now scaled up to consider auctions for procuring demand over multiple time slots ($m \geq 2$).

4.2.1. Daily unit costs

First, we assume that generators have a single daily unit cost for all time slots. As show in the experiments in Appendix B.2, the optimal auction corresponds to the mechanism outlined in Proposition 2.7 applied for each slot. Since problem (1) is separable in each time slot j, generators do not have intertemporal incentives, and the asymmetry in capacity or demand between slots does not influence the daily expected cost. Instead, the aggregate demand and capacity throughout the day are more relevant in determining the expected payment levels to generators.

Takeaway 4. *The optimal auction in multi-time slot settings with equal unit costs across slots aligns with applying the mechanism outlined in Proposition 2.7 separately for each time slot.*

4.2.2. Time slot unit costs

Next we consider settings where generators have different unit costs for each time slot. In this set of experiments, there are 2 generators competing for procuring demand during 2 time slots, and unit costs are drawn independently from $U[0,1]$ across both generators and time slots. We consider the following settings:

Q 2 generators with no capacity constraints in both time slots ($\bar{q}_{ij} \geq 0.5$). Demand is 0.5 during each time slot.

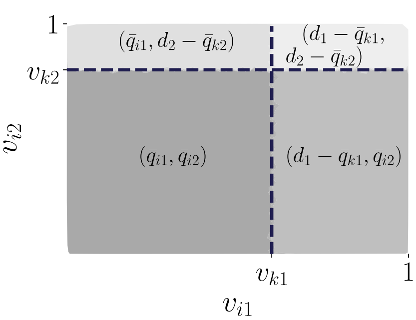
R 2 generators: the first with a capacity of 0.3 and the second with a capacity of 0.4 in each slot. Demand is 0.5 during each time slot.

S 2 generators: the first with a capacity of 0.3 and the second with a capacity of 0.4 in each slot. Demand is 0.6 during the first slot and 0.4 during the second.

T 2 generators: the first with a capacity of 0.2 in the first slot and 0.4 in the second slot. The second generator has a capacity of 0.4 in each slot. Demand is 0.5 in each time slot.

In this set of experiments, electricity units are considered to be different across time slots, thus the design problem blends elements of both multi-unit (inside each time slot) and multi-item auctions (between time slots). To better understand the following results, let us define two different mechanisms. First, the *independent* auction (6) sequentially allocates units with the lowest costs within each time slot until capacity constraints or demand are met. Thus the allocation rule $\boldsymbol{g}_i(\boldsymbol{v})$ can be visualized as

$$\boldsymbol{g}_i(\boldsymbol{v}) =$$

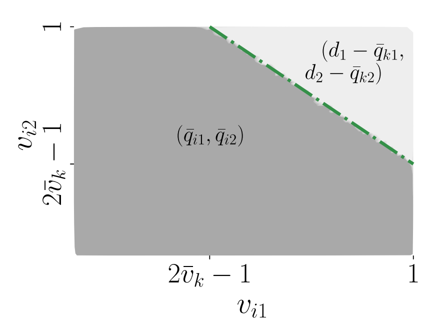


, , (6)

for each generator $i \in \{1,2\}, i \neq k$. The auction is separable and coincides with mechanism (5) from Proposition 2.7 (following Iyengar and Kumar (2008) and Chaturvedi (2015)) applied during each time slot.

In contrast, the *pure bundling* auction (7)— initially studied by Adams and Yellen (1976) and Palfrey (1983)—uses only the combined value of individual unit costs as the decision boundary of the allocation rule for each time slot, either by summing or averaging. This mechanism sequentially assigns at most the daily capacity to generators with the lowest combined value until demand is satisfied. As a result, the probability of dispatching each unit in one time slot decreases with the value of the unit cost from the other time slot. Thus the allocation rule $\boldsymbol{g}_i(\boldsymbol{v})$ in the pure bundling case can be visualized as

$$\boldsymbol{g}_i(\boldsymbol{v}) =$$



, , (7)

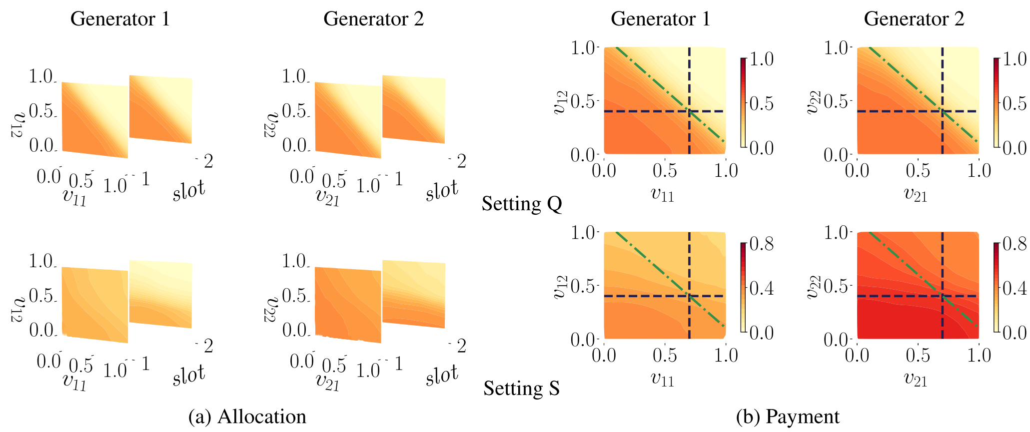


Fig. 14. Partial plot of the auction for settings Q and S. Generators' rules are displayed as functions of their unit costs in each slot, with costs fixed for the other generator (0.7 in slot 1 and 0.4 in slot 2). Left block: allocation rule during each time slot and during the day. Right block: payment rule. Blue dashed and green dashed–dotted lines represent the decision boundaries for the independent and the pure bundling auction, respectively.

Table 4
Summary of the performance on the test set for settings Q–T. $\bar{\boldsymbol{q}} = ((\bar{q}_{11}, \bar{q}_{21}), (\bar{q}_{12}, \bar{q}_{22}))$ denotes the capacity vector, $\boldsymbol{d} = (d_1, d_2)$ the demand vector, m the number of slots, and h the number of bids per day.

	$\bar{q}$	d	m	h	*RegretNet*		
					cost	*rgt*	*ccp*
Q	*Uncapacitated*	(0.5, 0.5)	2	2	0.6241	<0.001	–
R	((0.3, 0.4), (0.3, 0.4))	(0.5, 0.5)	2	2	0.8478	<0.001	<0.001
S	((0.3, 0.4), (0.3, 0.4))	(0.6, 0.4)	2	2	0.8648	<0.001	<0.001
T	((0.2, 0.4), (0.4, 0.4))	(0.5, 0.5)	2	2	0.8556	<0.001	<0.001

for each generator $i \in \{1, 2\}, i \neq k$, where $\bar{v}_k = (v_{k1} + v_{k2})/2$. Here the payments are determined by the area under the allocation rule. The combination of the independent and the pure bundling auctions leads to a so-called *mixed bundling* auction (Adams and Yellen, 1976).

In practice, some electricity markets, such as the Colombian market, restrict generators to a single daily bid to mitigate market power, usually representing the average cost across all time slots (CREG, 2009). Note that if the optimal auction reduces to a function solely of the average unit costs (such as mechanism (7)), then, using a single daily bid representing the average unit cost is also optimal.

Table 4 illustrates the test set performance of the learned auctions for settings Q to T, where we observe low constraint violations (≤0.001). In settings Q and R, the average costs are lower compared to those of the independent auction (6), which result in expected costs levels of 0.6664 and 0.8665, respectively. Fig. 14 shows the partial plot of the auction rules for each generator, keeping the unit costs constant for the other bidder, in setting Q (top row) (refer to Figure B8 in the Appendix for setting R). Observe that the learned rules have a decision boundary more similar to that of the pure bundling auction (7). One explanation for the high degree of bundling is that using a signal (the combined value) with lower variance, increases competition and reduces the information rents paid (Bakos and Brynjolfsson, 1999; Schmalensee, 1984). However, for extreme opposing unit cost realizations, it becomes optimal to reduce bundling to increase efficiency (i.e., assigning more units to the generator with significantly lower unit costs), resulting in the optimal design being a mixed bundling auction. For instance, when generator 2 has a unit cost of 0 in the first slot and 0.8 in the second, while the pure bundling auction assigns it 0.5 units (since $v_{11} + v_{12} > v_{21} + v_{22}$), the learned auction assigns it fewer units in the second slot (this is even more evident in setting R).

For settings where demand and capacities are asymmetric across time slots (settings S and T), Table 4 shows that the learned auction has expected costs more similar to those of the independent auction (6) (with an expected cost of 0.8665, same as C). In Fig. 14 (bottom row), it is more evident that the decision boundary of the pure bundling auction inadequately separates the regions of the learned auction, better resembling the independent auction (6) (for setting T see Figure B8 in the Appendix). This is due in part to high levels of bundling not meeting the *incentive compatibility* (IC) condition when allocated units vary between slots. In these instances, the cost function no longer depends solely on the combined values and by bidding higher average unit costs, generators can reduce their production costs and increase profits by procuring fewer units in high unit cost slots, even if they receive lower payments. Nonetheless, some level of bundling is still optimal, resulting in the optimal design being a mixed bundling auction with less bundling than in the first two settings presented in this section (Q and R).

These results are in line with those in Armstrong (2000) for multi-item auctions, where it is shown that when valuations are sufficiently correlated, the optimal design is the independent auction, while for independent valuations, some form of bundling is optimal. Moreover, our results further indicate that when demand and capacity constraints vary between time slots, the ability of the auction to bundle is reduced. This suggests that unless unit costs are equal across time slots (for which pure bundling and the independent auctions yield the same expected cost) limiting the number of bids (e.g., allowing only a single daily bid representing average costs) is inefficient, especially if demand and capacities differ across time slots. Conversely, insufficient bundling is also inefficient, particularly when unit costs are uncorrelated and demand and capacities are symmetric.

Takeaway 5. *In the previous multi-time slot settings with varying unit costs between slots, the optimal design is a mixed bundling auction, with the optimal bundling level constrained by the demand and capacity asymmetry across time slots.*

4.3. Real data experiments

This section considers experiments using real data from the Colombian wholesale electricity market throughout 2022. We assume that generators submit a single daily unit price bid, as it is custom in this market, and a linear generator cost function, for simplicity.

Market information for Colombian power generation plants in 2022 was collected at the plant level, including daily unit prices, hourly declared capacities and consumer demand sourced from the Colombian system operator, XM. Plants were classified according to their energy source and then grouped into 5 representative generators: liquid-fueled thermal, gas/coal thermal, hydro, wind, and solar. Unit costs were estimated using a generator-specific approach to reflect their distinct cost structures. For thermal plants unit costs were calculated using an engineering formula based on fuel prices, heat rates, and variable and maintenance costs (Mansur, 2008). For hydro plants, we estimated opportunity costs from bids and marginal dispatch prices from XM, following Riascos et al. (2016). For wind and solar generators, we relied on the levelized cost of electricity (LCOE) benchmarks (IRENA, 2022). Capacities for thermal and hydro plants were taken from declared hourly availability, while wind and solar capacities were constructed using wind speed data (IDEAM) and solar irradiance data (Nasa), combined with standard turbine and PV panel models, and equations for electricity output (Chedid et al., 1998; Hetzer et al., 2008).

In addition, we consider only 2 time slots: the first corresponding to low-demand hours (between 10 pm and 8 am) and the second to high-demand hours (between 9 am and 9 pm). Hourly data were aggregated at each time slot by averaging unit costs and summing capacities and demand. The data were then normalized by dividing the costs and quantities by the maximum observed unit cost and demand values, respectively. Finally, distributions were fitted to the normalized data for each generator and time slot.

In the experiments below, capacities and demand are modeled as random variables, with their realizations regarded as non-strategic and not private information (see[2]). Unit costs are independently drawn across generators and equal across slots, with each generator submitting a single daily unit-price bid. Further details on the estimation of the model parameters are available in Appendix A.3.

In this section, we start with a baseline setting B0, where only thermal and hydro generators compete in the auction. Then, we increase the average aggregate daily capacity of the initial setting by 10%, 20%, and 30% through the integration of wind and solar power (experiments R10–R30), and by expanding hydropower (experiments H10–H30).

Baseline.

B0 3 generators: liquid-fueled thermal, gas/coal fueled thermal and hydro generator.

Wind–solar integration.

R10 5 generators: aggregate capacity is expanded by 10% on average by introducing a wind and a solar generator.
R20 5 generators: aggregate capacity is expanded by 20% on average by introducing a wind and a solar generator.
R30 5 generators: aggregate capacity is expanded by 30% on average by introducing a wind and a solar generator.

Hydropower expansion.

H10 5 generators: aggregate capacity is expanded by 10% on average by introducing 2 additional hydro generators.
H20 5 generators: aggregate capacity is expanded by 20% on average by introducing 2 additional hydro generators.
H30 5 generators: aggregate capacity is expanded by 30% on average by introducing 2 additional hydro generators.

The unit cost distributions by source are shown in Fig. 15. The liquid-fueled thermal generator exhibits the highest unit costs and the wind generator the lowest. The unit cost distribution of the hydro generator is skewed to the right, indicating instances of decreased rainfall. Nonetheless, on average, hydro, wind and solar generators have similar unit costs. Fig. 16 shows the average capacities for each generator for settings R10–R30, where we introduce wind and solar power to expand the aggregate capacity at each level. Note that the hydro generator has the highest capacity, reflecting Colombia's reliance on hydropower. Moreover, the increase in capacity is not uniform across time slots, especially due to lower solar power generation during the first time slot, where sunlight is not available.[12] For experiments that introduce the additional hydro generators, they share the same unit cost and capacity distributions as those in the baseline setting, but the capacity distribution is rescaled to expand aggregate capacity at each level.

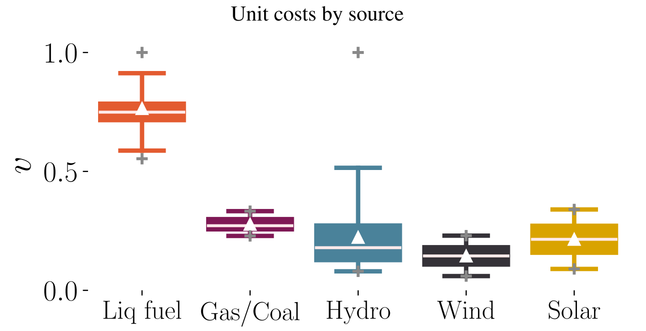


Fig. 15. Boxplot of unit costs for each generator type. △ and + markers indicate mean and max/min values, respectively.

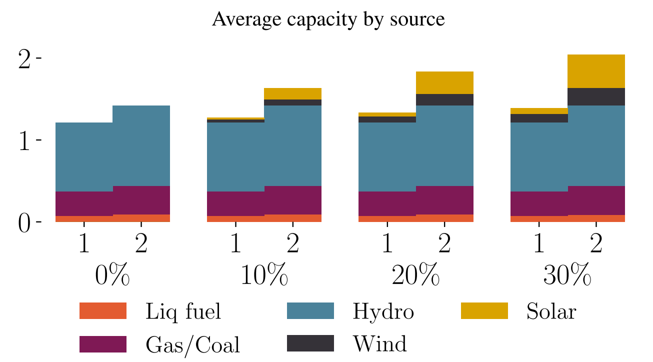


Fig. 16. Average capacity for each generator during each time slot $j \in \{1, 2\}$ for settings B0 and R10–R30.

Table 5 illustrates the test set performance, revealing low constraint violations (≤0.001). As seen in previous experiments, significant reductions in expected costs occur with expansions in aggregate capacity and number of generators, regardless of whether the increase stems from wind and solar energy, or from hydropower. However, integrating wind and solar energy results in slightly larger cost reductions of 1.1%, 3.3% and 5.1%, compared to settings that introduce additional hydro generators at 10%, 20%, and 30% expansion, respectively (though these magnitudes may be partially due to approximation errors). This improvement is more significant at the 95th percentile, with reductions of 8.8%, 8.2%, and 16.1% in generation costs, respectively.

For settings R10–R30, Fig. 17 shows that increasing wind and solar power integration reduces the reliance on liquid-fueled thermal generation, especially during the second time slot, where wind and solar generators have greater capacity. Moreover, the payment for the hydro generator decreases considerably, which accounts for most of the decline in expected costs. Observe that this is mainly due to the reduction in the information rent, reflecting a decrease in its markup. Since wind and solar power have similar unit costs on average, the integration increases competition, lowering the opportunity cost of misreporting for the hydro generator. In practice, these findings suggest that policies aimed at expanding aggregate capacity should encourage the adoption of competitive and cost-effective generation technologies and the entry of new generators.

[12] In addition, capacity was increased using 40% wind power and 60% solar power.

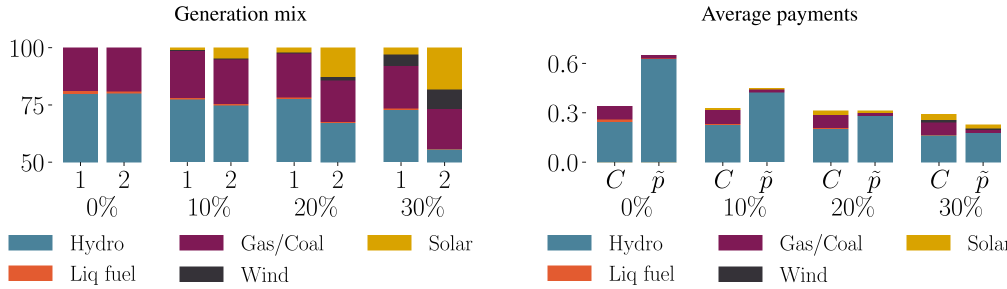


Fig. 17. Generation mix (left) and average payments (right) for each generator across settings B0 and R10–R30. For each time slot $j \in \{1,2\}$, the generation mix represents the participation of each generator in the total allocation (in %). Payments are broken down into the costs incurred by generators C_i, and the information rent $\tilde{p}_i$.

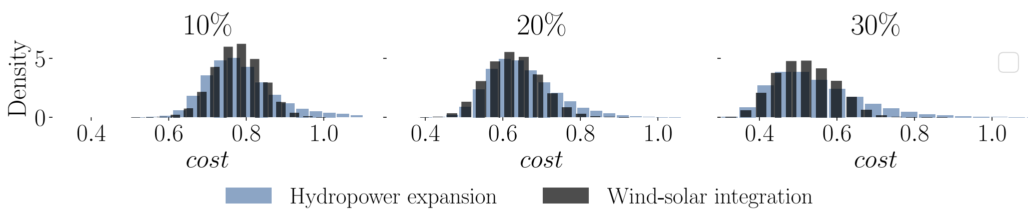


Fig. 18. Distribution of generation costs across capacity expansion levels (10%, 20%, and 30%) for settings R10–R30 in black bars and for H10–H30 in blue bars.

Table 5
Summary of the performance in the test set for settings B0–H30. *p*5 and *p*95 represent the 5th and 95th percentiles of the generation cost in the learned auction, respectively.

	Capacity expansion	*RegretNet*				
		cost	*p5*	*p95*	*rgt*	*ccp*
Baseline						
B0	0%	0.9470	0.8673	1.1170	<0.001	<0.001
Wind - solar integration						
R10	10%	0.7771	0.6788	0.8774	<0.001	<0.001
R20	20%	0.6265	0.5169	0.7421	<0.001	<0.001
R30	30%	0.5178	0.4016	0.6426	<0.001	<0.001
Hydropower expansion						
H10	10%	0.7855	0.6619	0.9618	<0.001	<0.001
H20	20%	0.6480	0.5282	0.8081	<0.001	<0.001
H30	30%	0.5458	0.3965	0.7656	<0.001	<0.001

Fig. 18 shows that integrating wind and solar generation is more effective in reducing the incidence of extreme generation cost levels compared to introducing new hydro generators. These elevated cost levels often coincide with extreme climate events, during which hydro generators incur high unit costs. In Colombia, where many reservoirs have low storage capacity relative to generation capacity, during periods of reduced rainfall, there is a higher risk that reserve capacity will deplete rapidly if hydro units actively generate electricity (Steeger et al., 2014; Carranza, 2015). As a result, the opportunity cost for hydro generators rises considerably, as they give up future payoffs due to low water resources. Integrating wind and solar power presents an alternative solution that increases competition and reduces costs on par with expanding hydropower. But more importantly, it has the additional effect of reducing the likelihood of extreme cost levels by complementing thermal generation during reduced rainfall periods.

These results are in line with the fact that hydropower tends to be a cost-effective source in Colombia, even compared to wind or solar power. These findings point out that in practice increasing competition for hydro generation could have a considerable impact on generation costs. Finally, some limitations of these experiments include: (1) computational constraints restrict the inclusion of all generators in the market and a more detailed time granularity, and (2) there may be approximation errors in the reported effects. Moreover, wind and solar energy generation tend to be more decentralized than hydro generation. Thus, comparing settings with an equal number of generators might not accurately reflect the impact of integrating wind and solar power. In addition, the use of the global levelized costs of electricity (LCOE) to estimate wind and solar unit costs may not fully account for different social costs and economic incentives in Colombia. Lastly, it would be better to use an underlying model of water inflows to estimate the opportunity costs of hydro generators, instead of relying on bids and prices of the current auction (see Pereira and Pinto (1985)).

Takeaway 6.

I. *For the Colombian case, we observe that expanding capacity and including more generators significantly reduces expected costs. This reduction is mainly due to the new generators having unit costs low enough to increase the competition for the initial hydro generator, thereby decreasing their information rents.*

II. *When compared to the settings that introduce additional hydro generators, integrating wind and solar power: (1) reduces the reliance on liquid-fueled thermal power; (2) is slightly more effective for reducing generation costs on average; (3) decreases the incidence of high generation costs, achieving considerable reductions of 8.8%, 8.2%, and 16.1% in the 95th percentile of generation costs across capacity expansion levels of 10%, 20%, and 30%, respectively.*

5. Conclusions

This work uses deep learning to find optimal strategy-proof mechanisms for electricity auctions in simplified settings. Experiments demonstrate that the proposed framework can recover known optimal solutions with accurate cost-level approximations (errors < 1 %) while incurring low constraint violations (≤0.001). Moreover, this study shows the framework's out-of-setting generalization ability, by discovering nearly optimal auctions for settings with uncertain capacity and demand, correlated generators, and multi-time slot demand procurement, achieving low constraint violations (≤0.001). In the case study using real data from the Colombian market, results indicate that integrating wind and solar power is slightly more effective for minimizing average costs compared to expanding hydropower. However, more notably, it decreases the likelihood of *high generation costs* during reduced rainfall periods, showing substantial reductions of 8.8%, 8.2%, and 16.1% in the 95th percentile of generation costs with capacity expansions of 10%, 20%, and 30%, respectively, when compared to increasing hydro generation.

This work hopes to open the door to explore further applications of this framework for electricity auctions. Future work can include constraints related to contamination levels or for ensuring fairness in the allocation (see, for example, Kuo et al. (2020)). It can also explore further scenarios with non-convex cost components and compare multi-part and block-bidding formats. Moreover, it could be valuable to assess the effectiveness of demand programs and reserve management and battery storage strategies for meeting a fluctuating demand, or dealing with the intermittent nature of renewables (as discussed in Mazidi et al. (2014) and Tang et al. (2020)). Some theoretical questions that this work leaves are: (1) characterizing the equilibrium for the uniform-price auction in asymmetric settings and comparing the generation cost against the optimal mechanism (see, for instance, Bichler et al. (2021)); and (2) assessing whether the designs are robust against collusion among generators. It would be useful to propose a method to find collusion-proof designs, especially for procurement auctions. Lastly, improvements in the neural network architecture involve incorporating the amortized misreport optimization of Rahme et al. (2020) for faster convergence.

CRediT authorship contribution statement

Valentina Cepeda: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Juan F. Pérez:** Writing – review & editing, Validation, Supervision, Methodology, Data curation, Conceptualization.

Acknowledgments

The authors would like to thank Prof. Álvaro J. Riascos, from Universidad de los Andes and Quantil, for his many valuable insights, encouragement and leadership in the inception and development of the project that led to the results presented in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eneco.2026.109176.

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